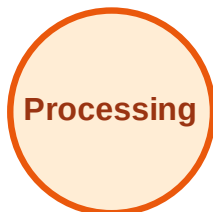


BFS Traversal

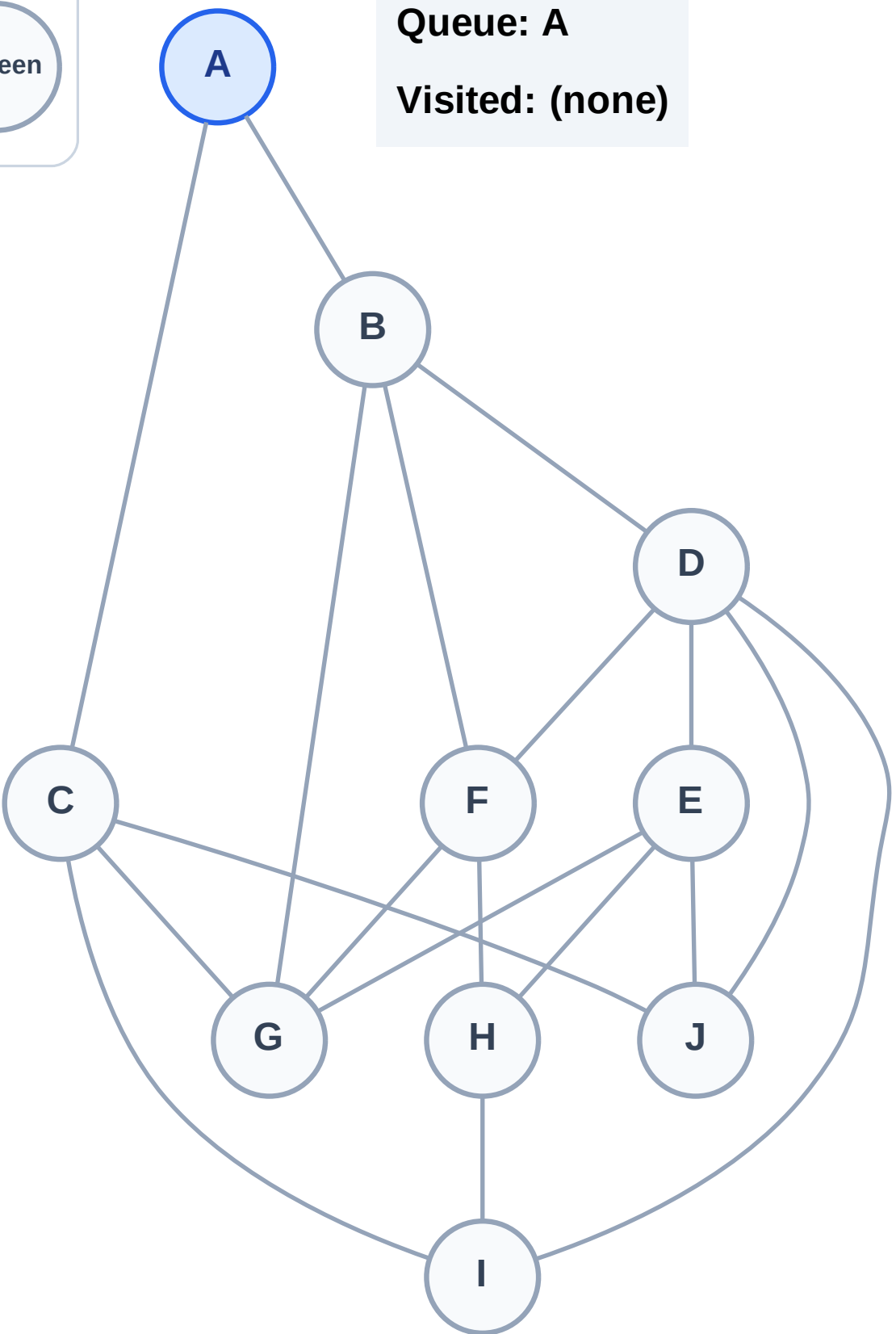
Step 1: Start at A

Legend



Queue: A

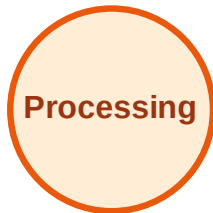
Visited: (none)



BFS Traversal

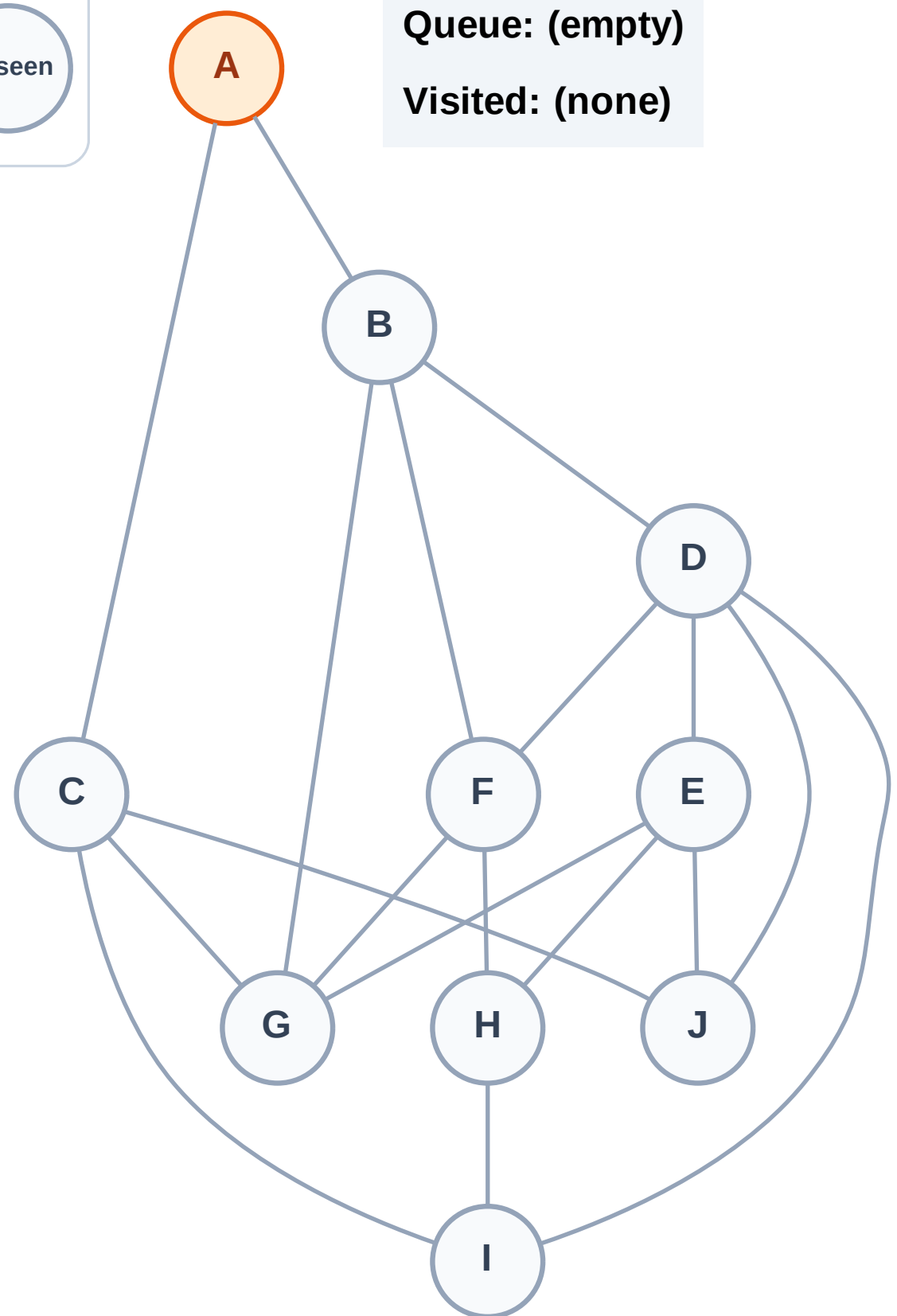
Step 2: Process node A

Legend



Queue: (empty)

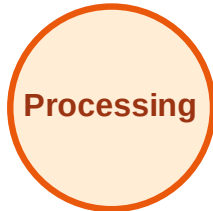
Visited: (none)



BFS Traversal

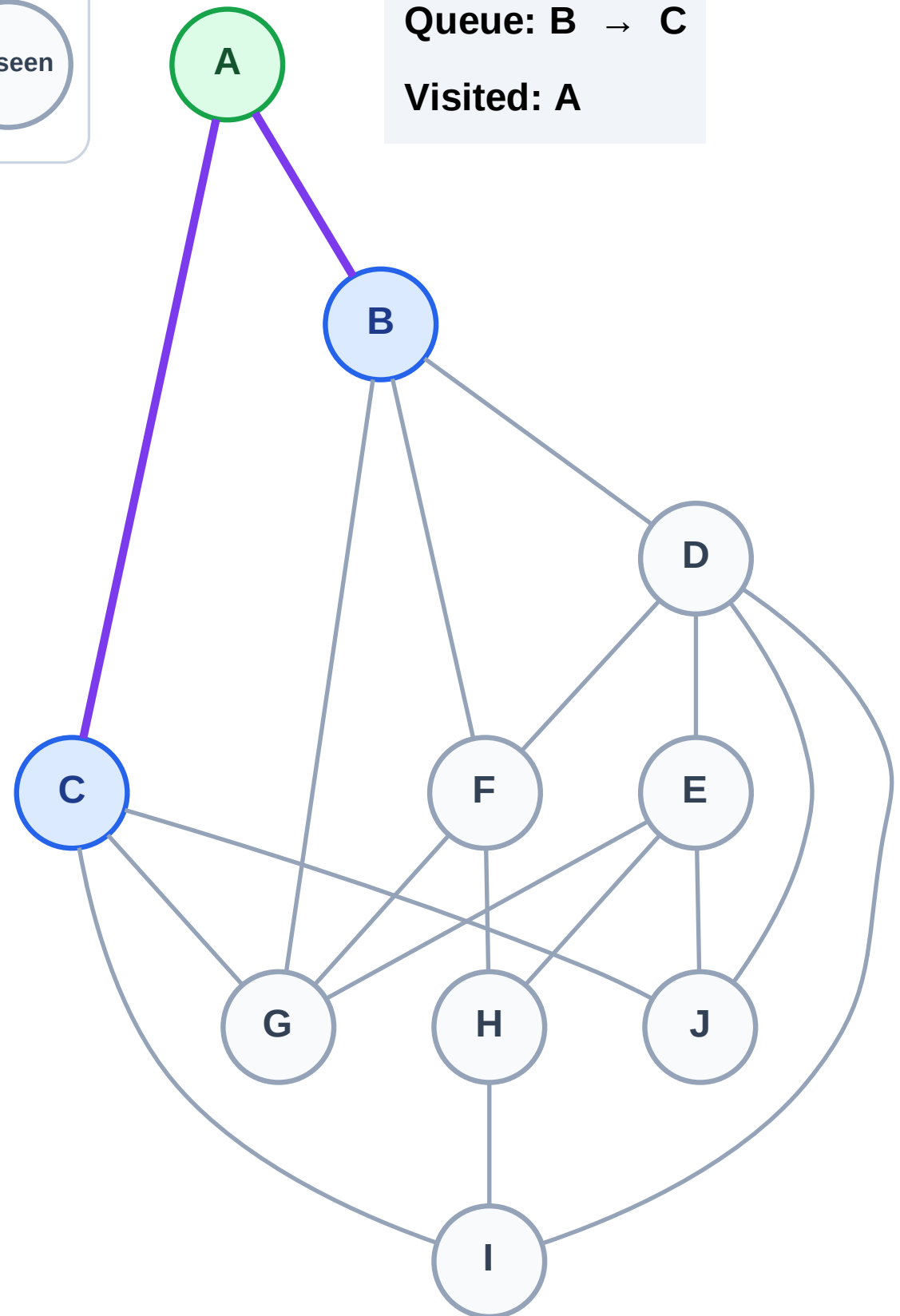
Step 3: Discover B, C from A

Legend



Queue: B → C

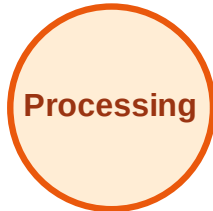
Visited: A



BFS Traversal

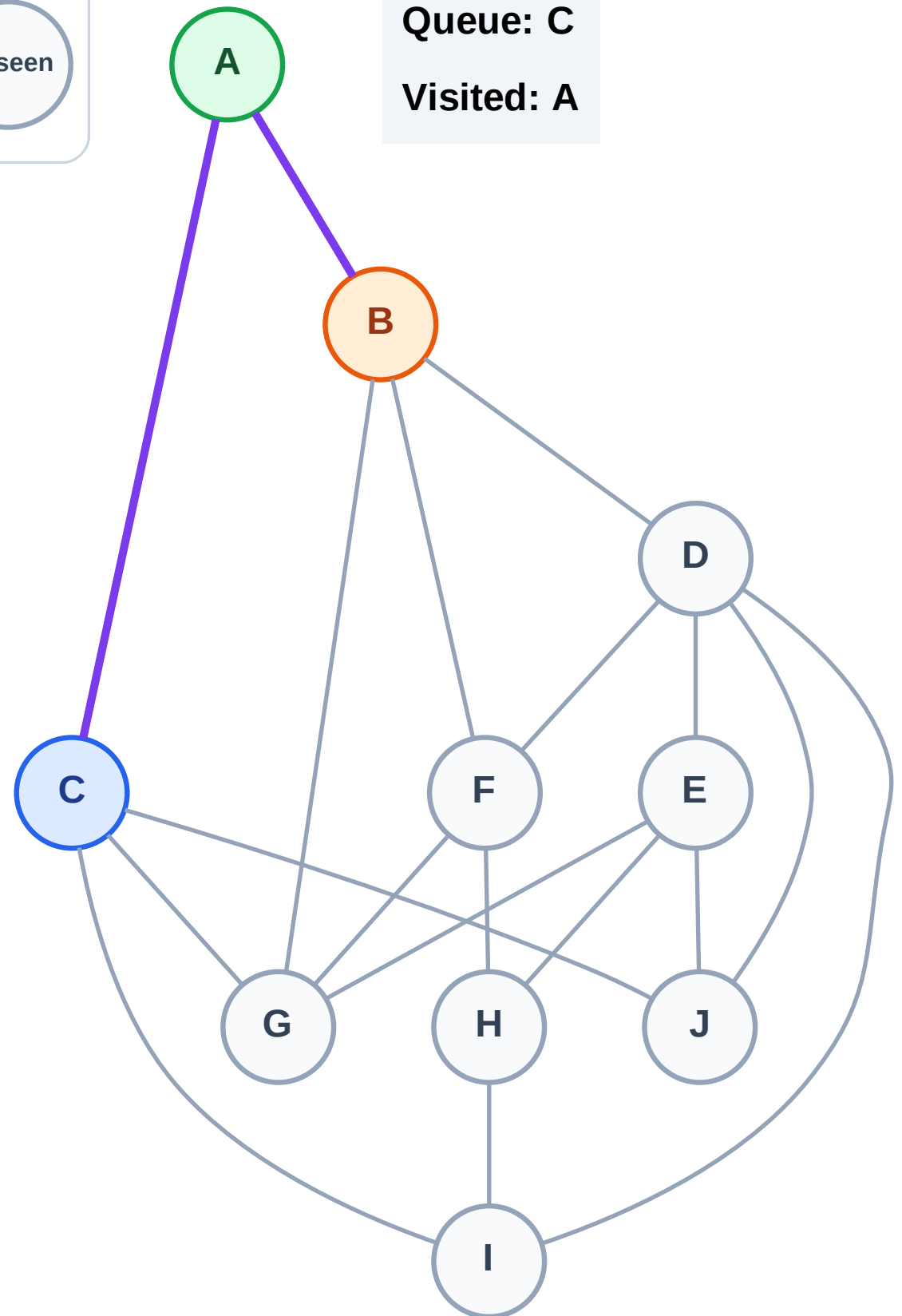
Step 4: Process node B

Legend



Queue: C

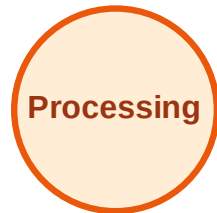
Visited: A



BFS Traversal

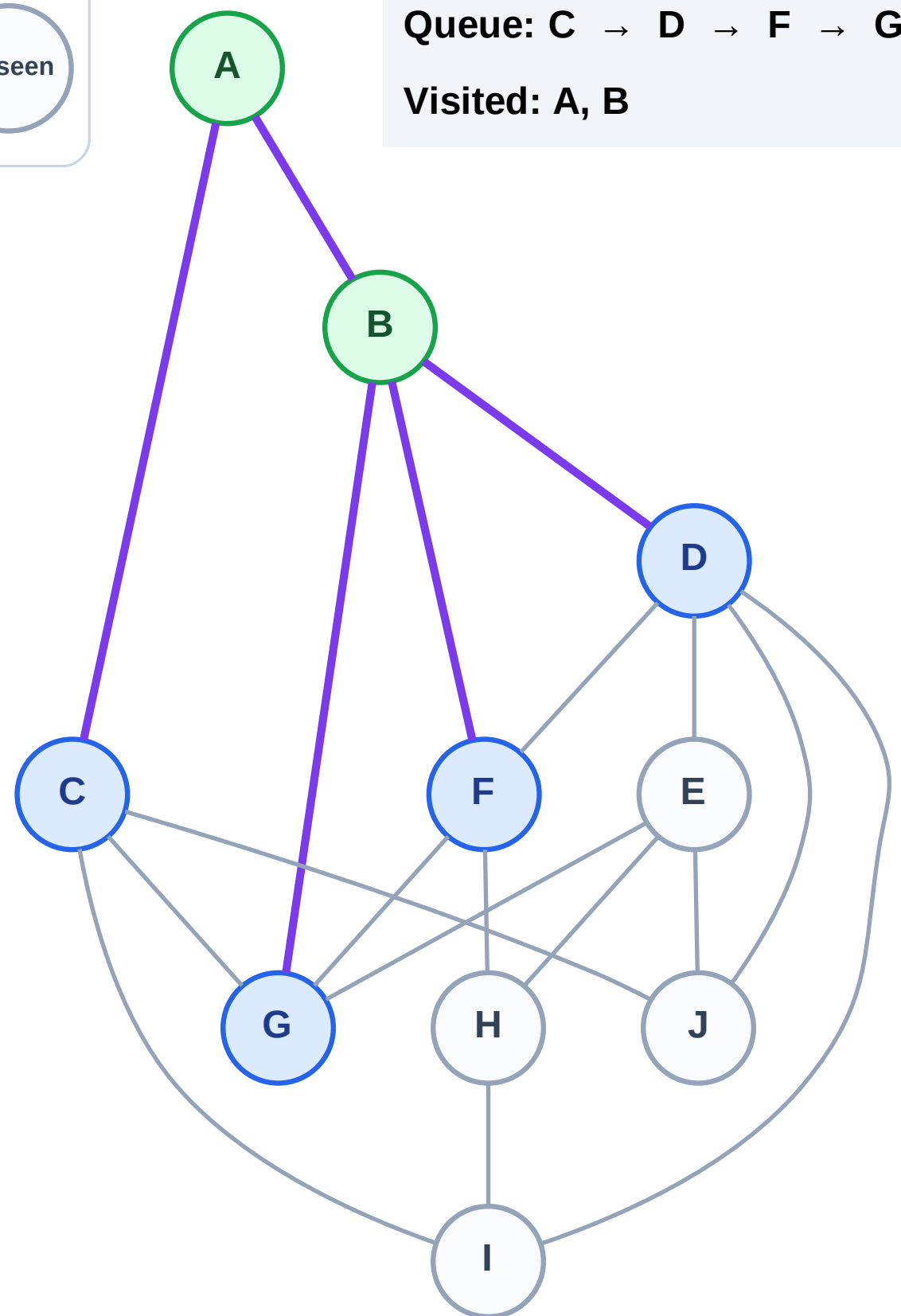
Step 5: Discover D, F, G from B

Legend



Queue: C → D → F → G

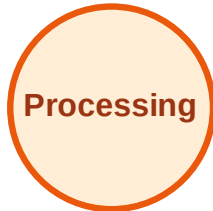
Visited: A, B



BFS Traversal

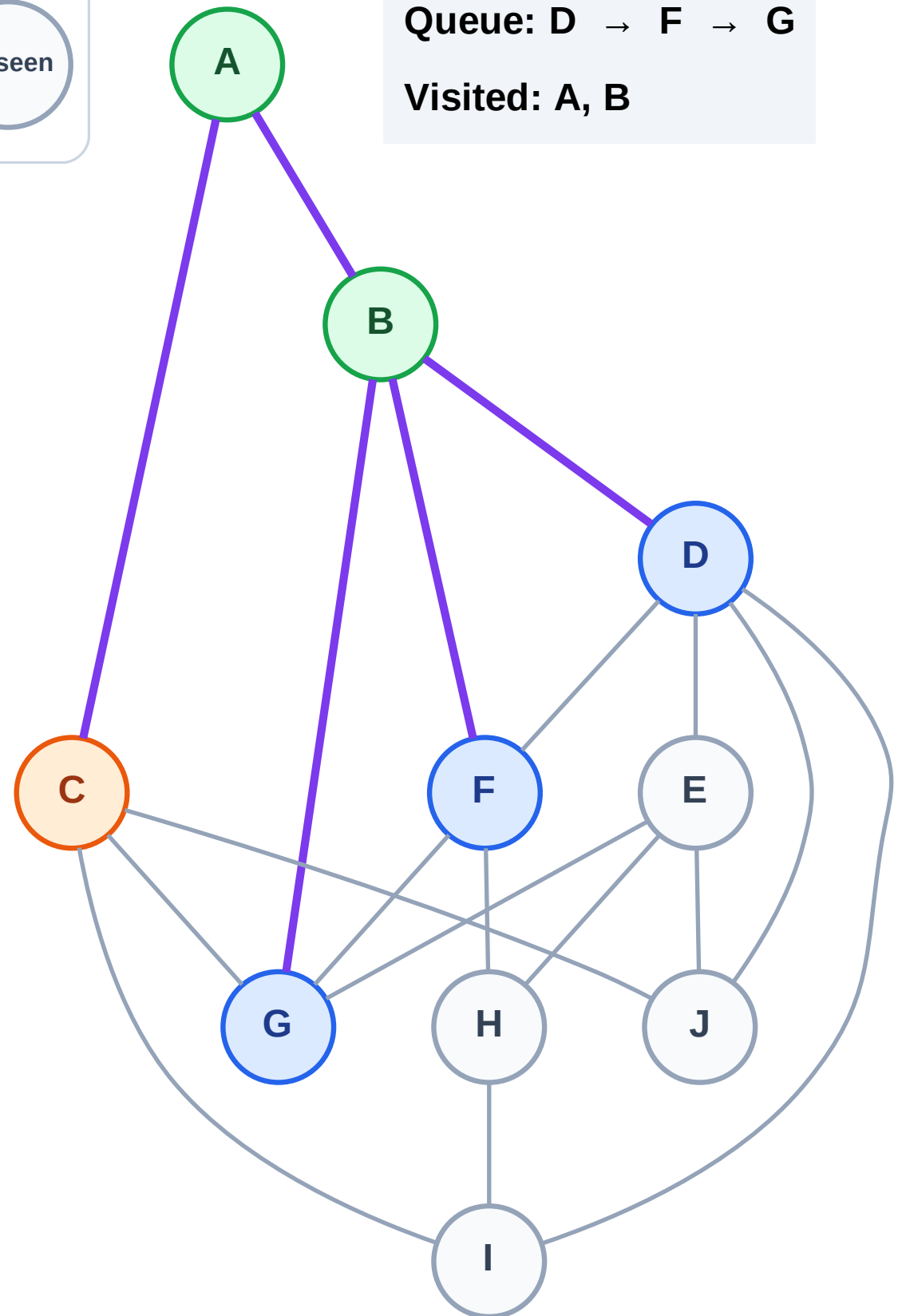
Step 6: Process node C

Legend



Queue: D → F → G

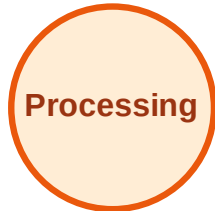
Visited: A, B



BFS Traversal

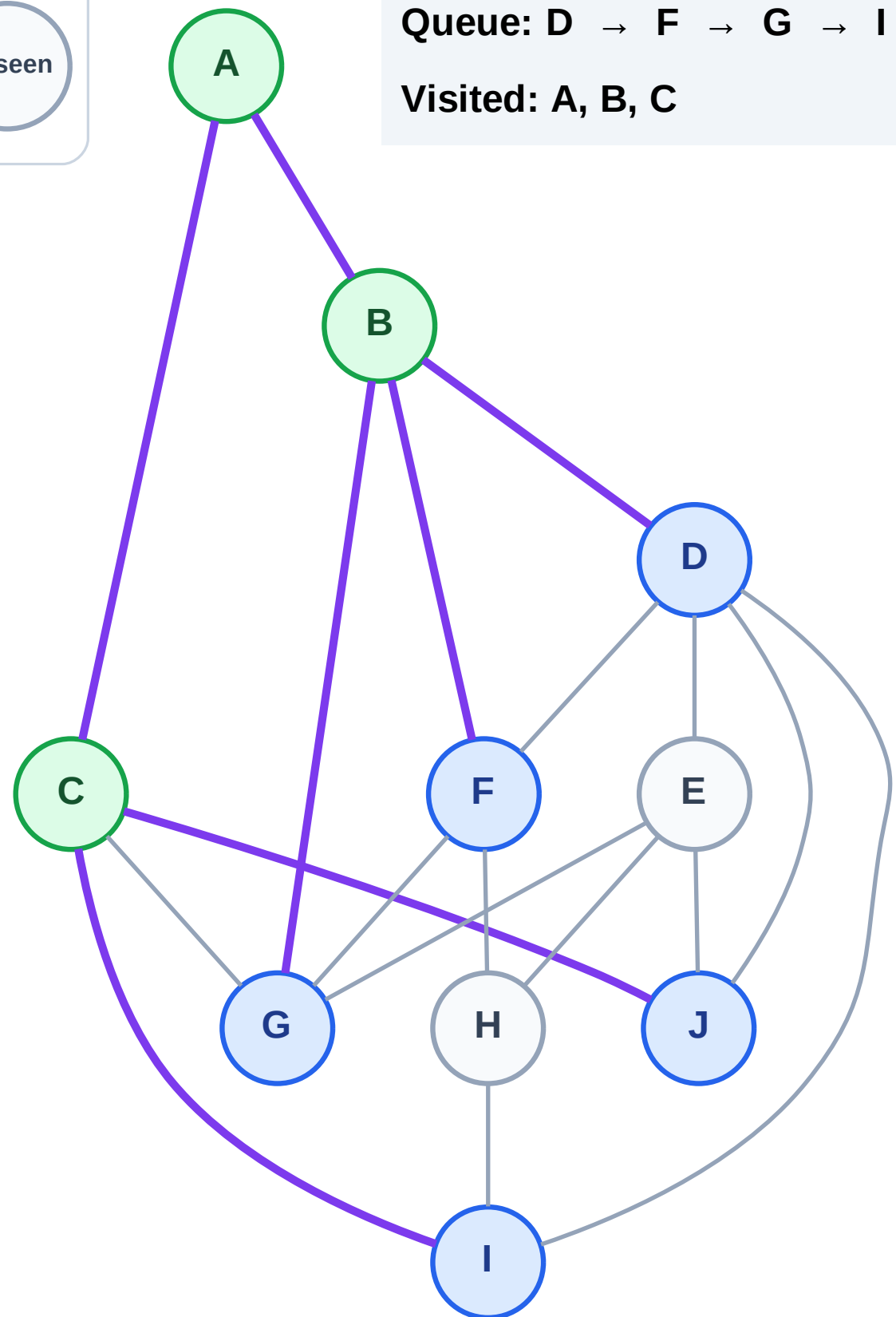
Step 7: Discover I, J from C

Legend



Queue: D → F → G → I → J

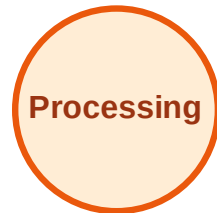
Visited: A, B, C



BFS Traversal

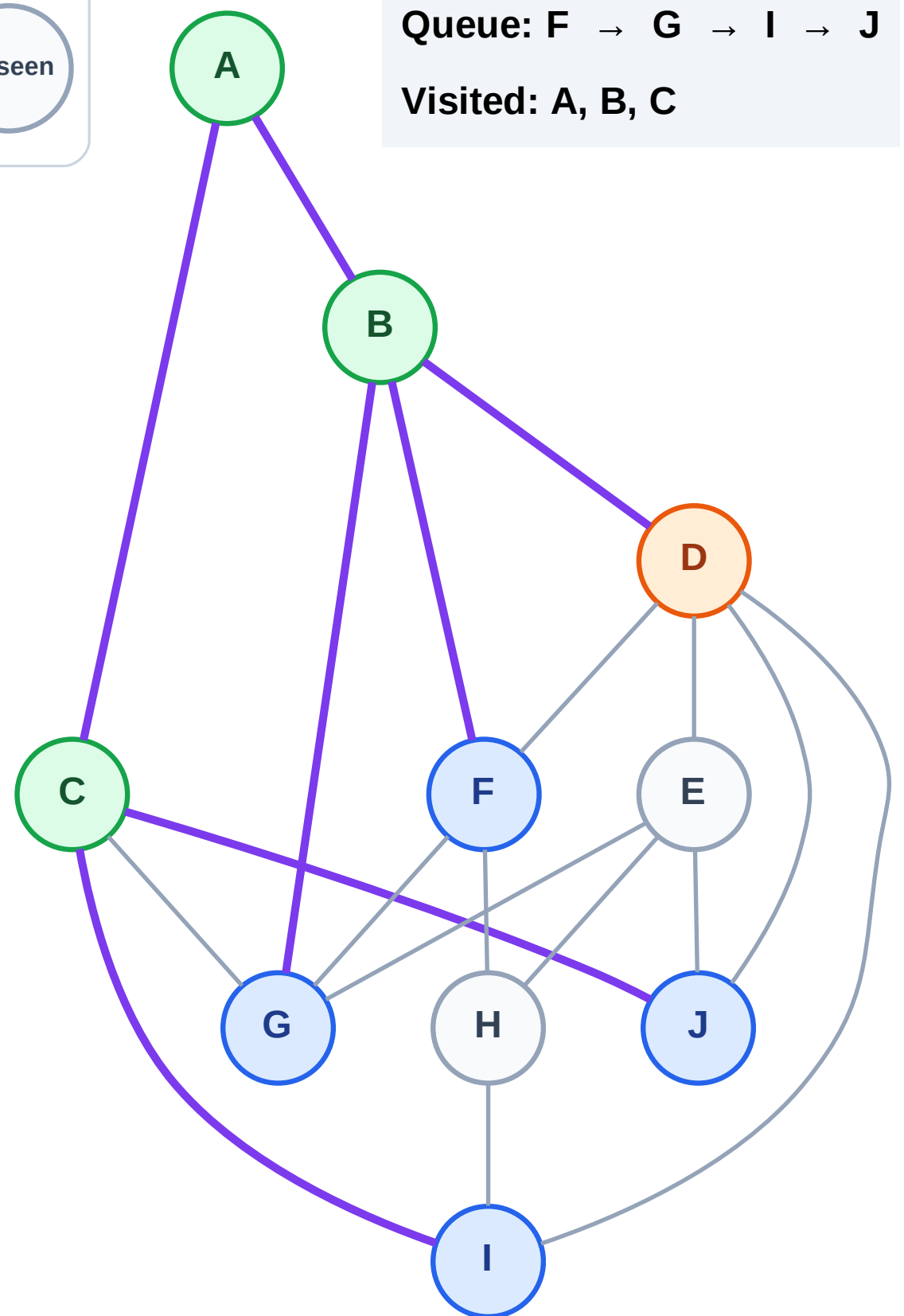
Step 8: Process node D

Legend



Queue: F → G → I → J

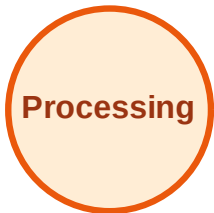
Visited: A, B, C



BFS Traversal

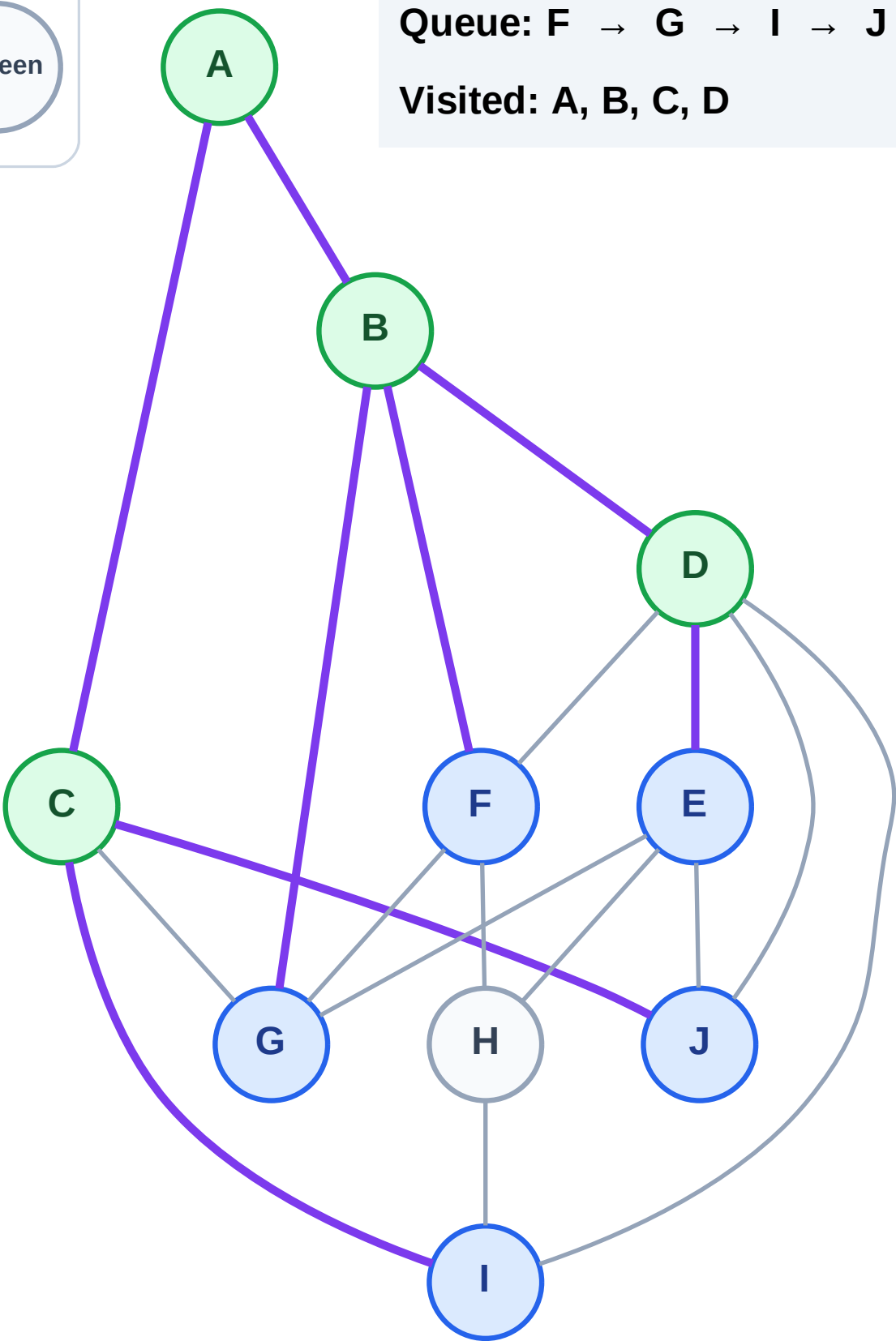
Step 9: Discover E from D

Legend



Queue: F → G → I → J → E

Visited: A, B, C, D



BFS Traversal

Step 10: Process node F

Legend

Complete

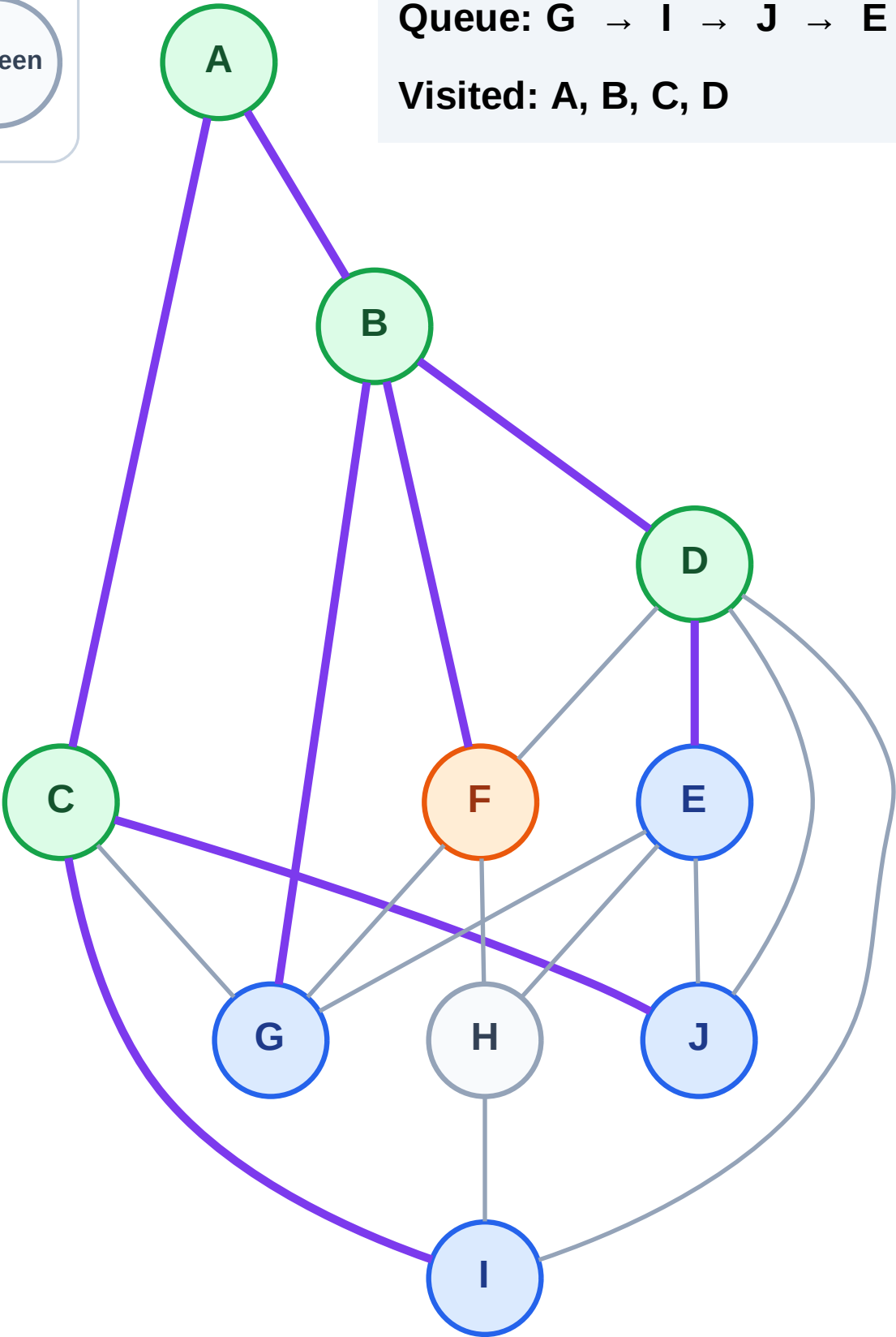
Processing

Discovered

Unseen

Queue: G → I → J → E

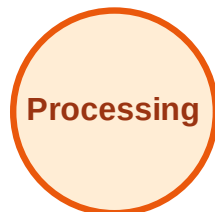
Visited: A, B, C, D



Step 11: Discover H from F

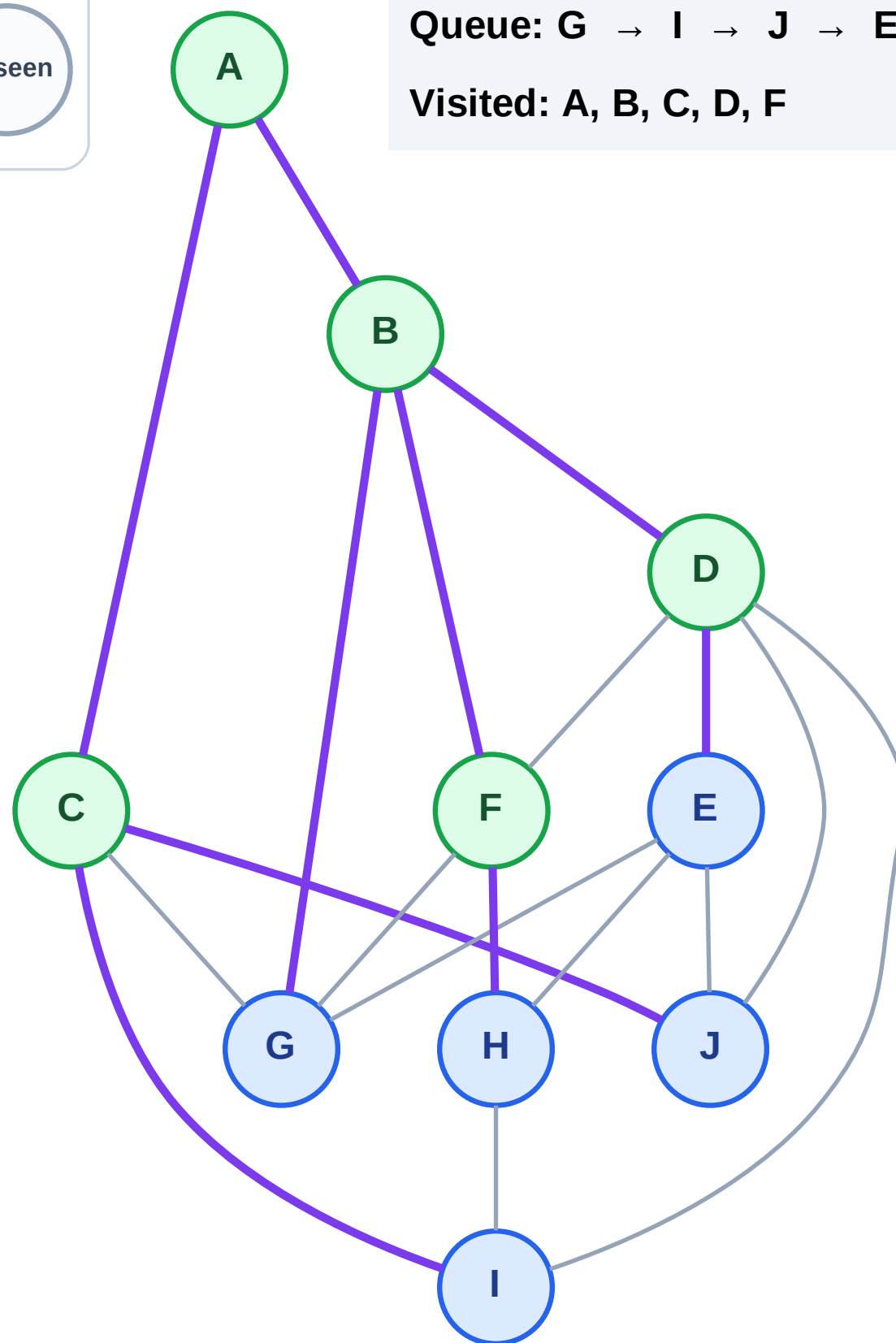
Step 11: Discover H from F

Legend



Queue: G → I → J → E → H

Visited: A, B, C, D, F



BFS Traversal

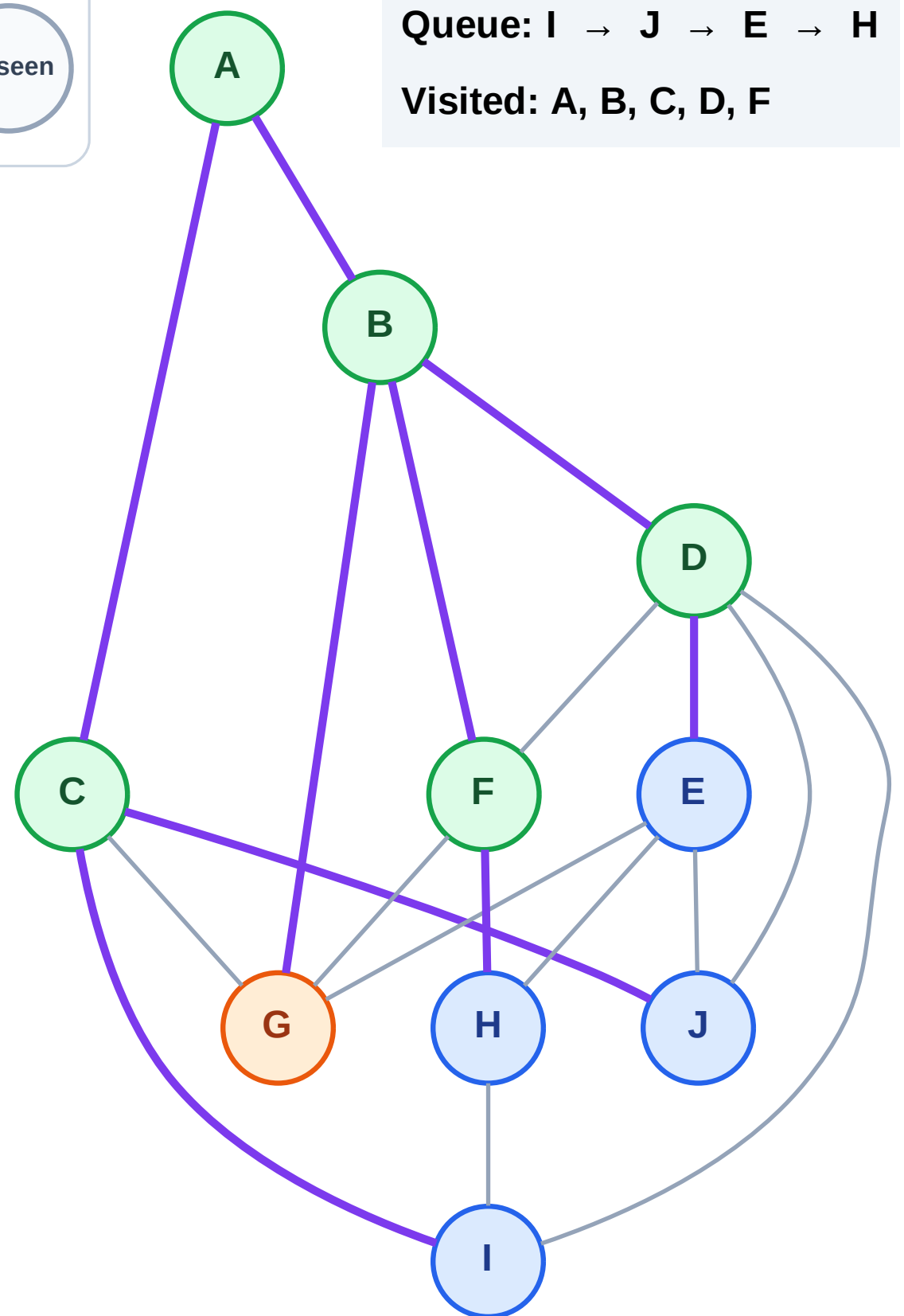
Step 12: Process node G

Legend



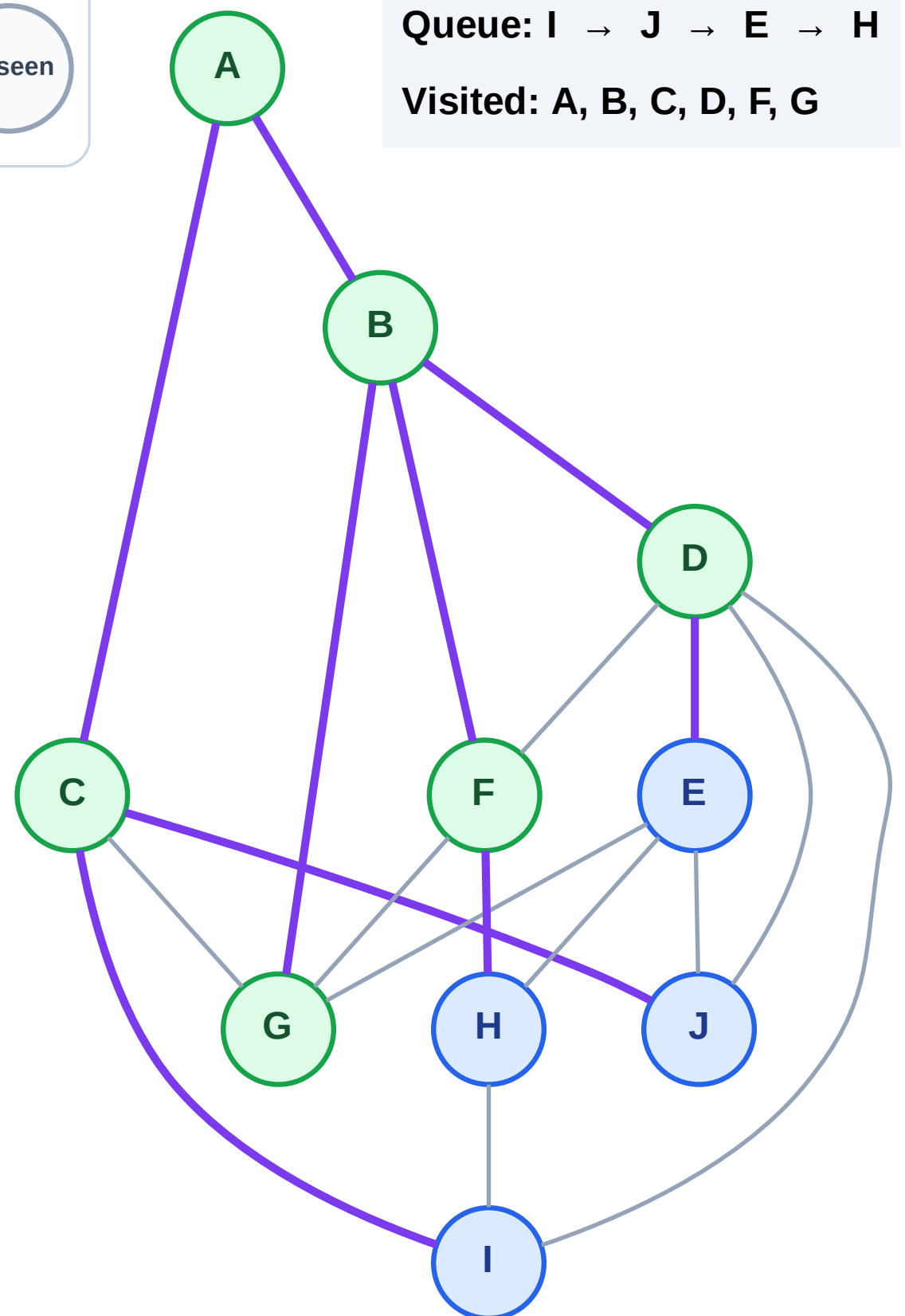
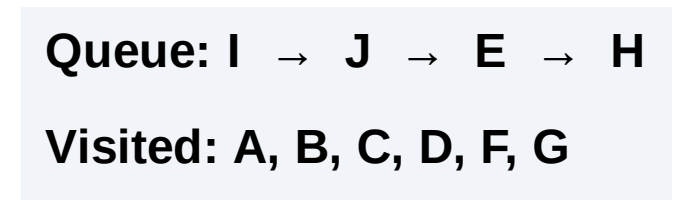
Queue: I → J → E → H

Visited: A, B, C, D, F



Step 13: No new neighbors from G

Step 13: No new neighbors from G



BFS Traversal

Step 14: Process node I

Legend

Complete

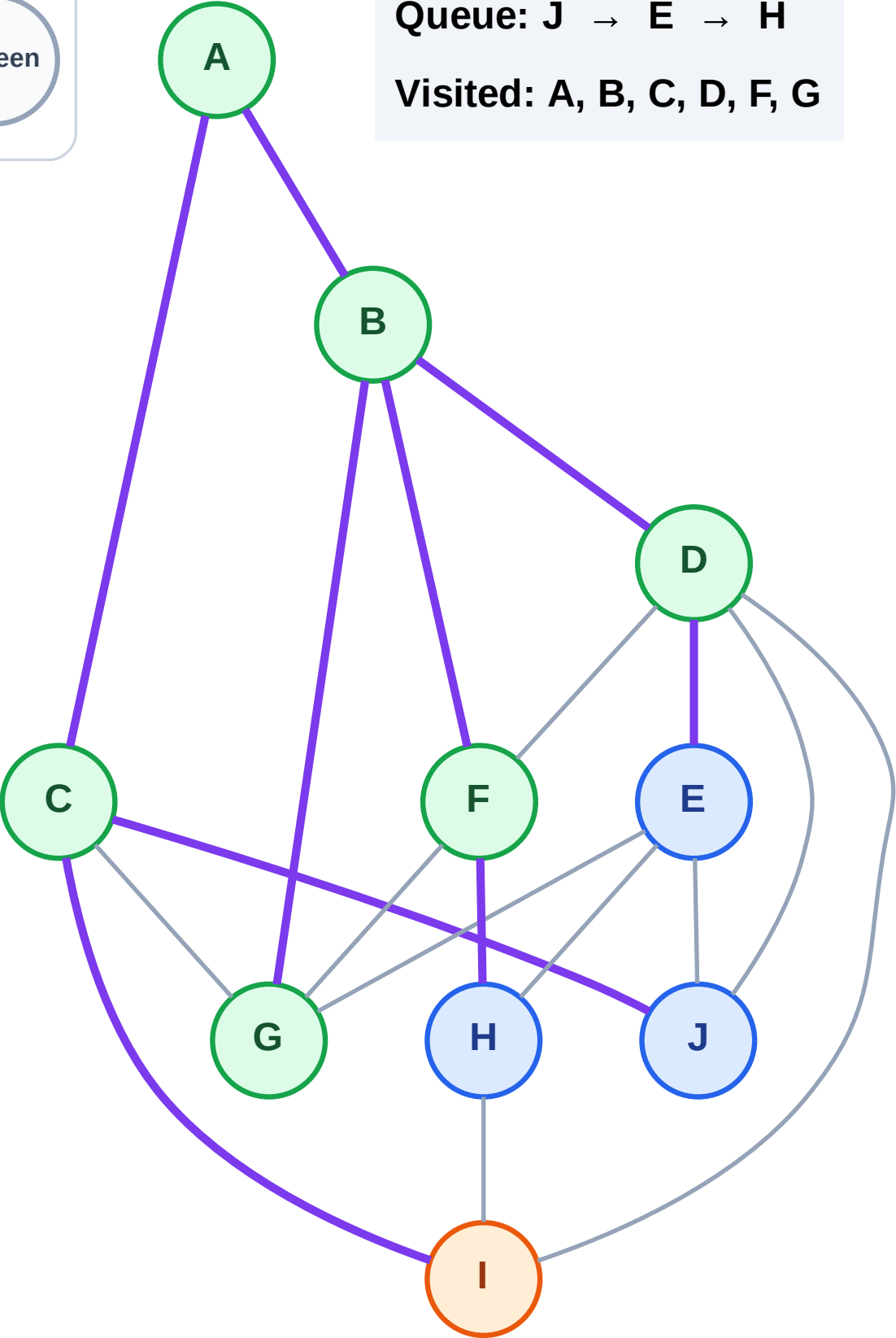
Processing

Discovered

Unseen

Queue: J → E → H

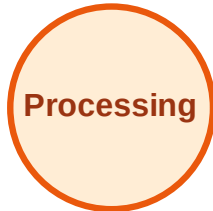
Visited: A, B, C, D, F, G



BFS Traversal

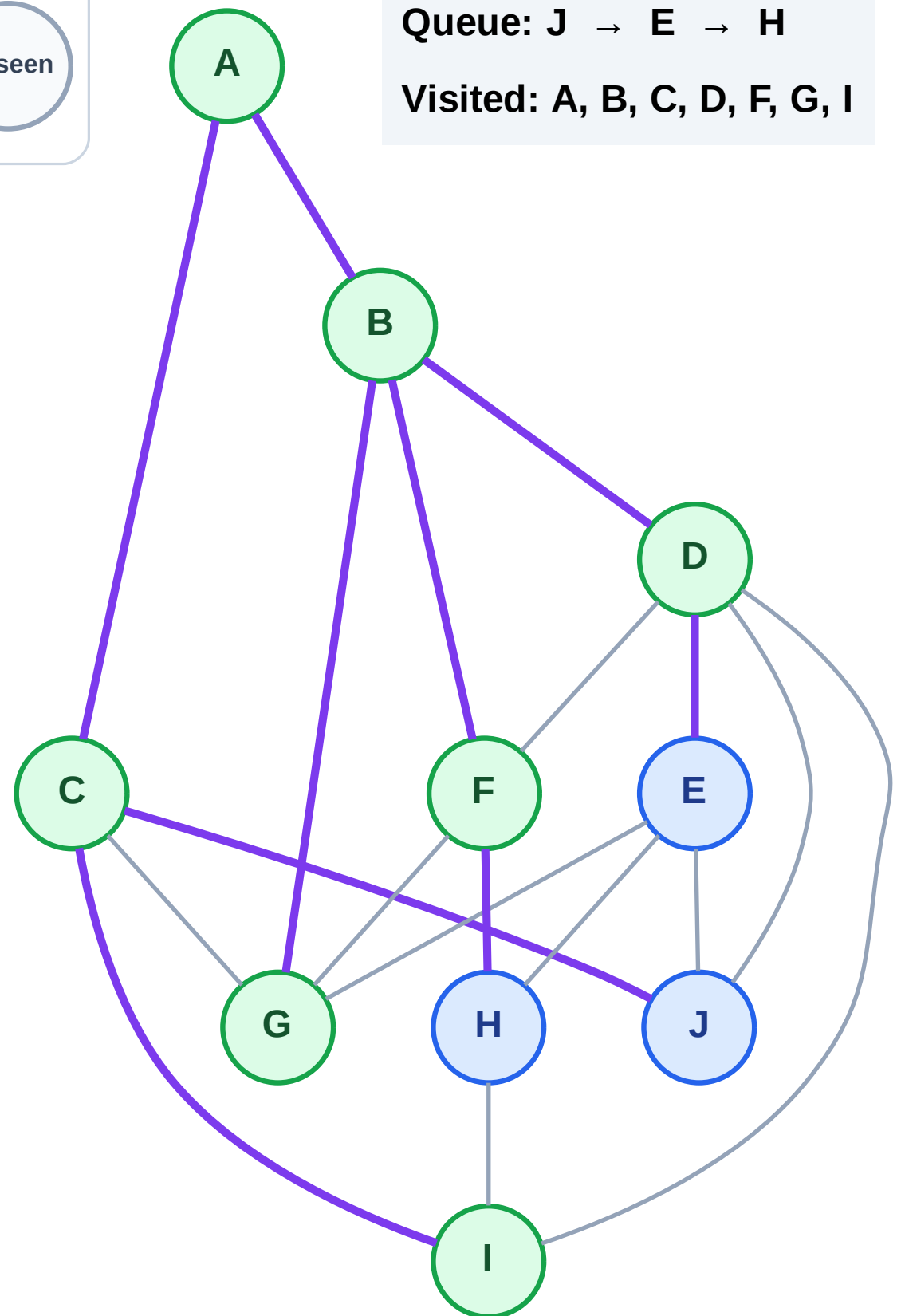
Step 15: No new neighbors from I

Legend



Queue: J → E → H

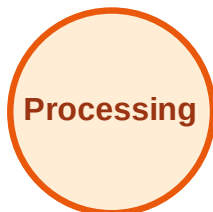
Visited: A, B, C, D, F, G, I



BFS Traversal

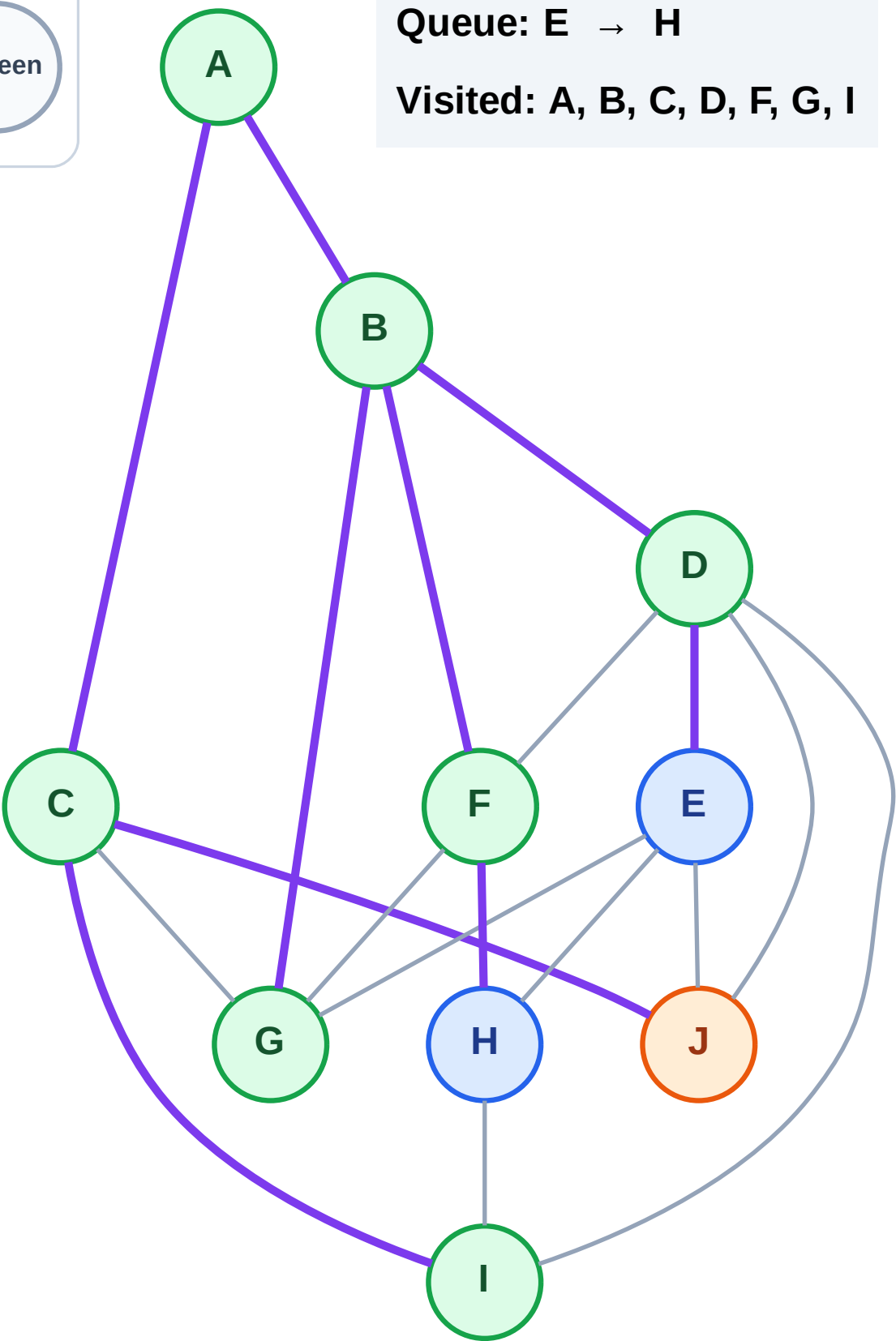
Step 16: Process node J

Legend



Queue: E → H

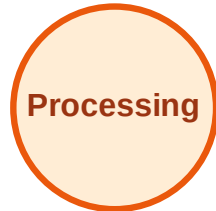
Visited: A, B, C, D, F, G, I



BFS Traversal

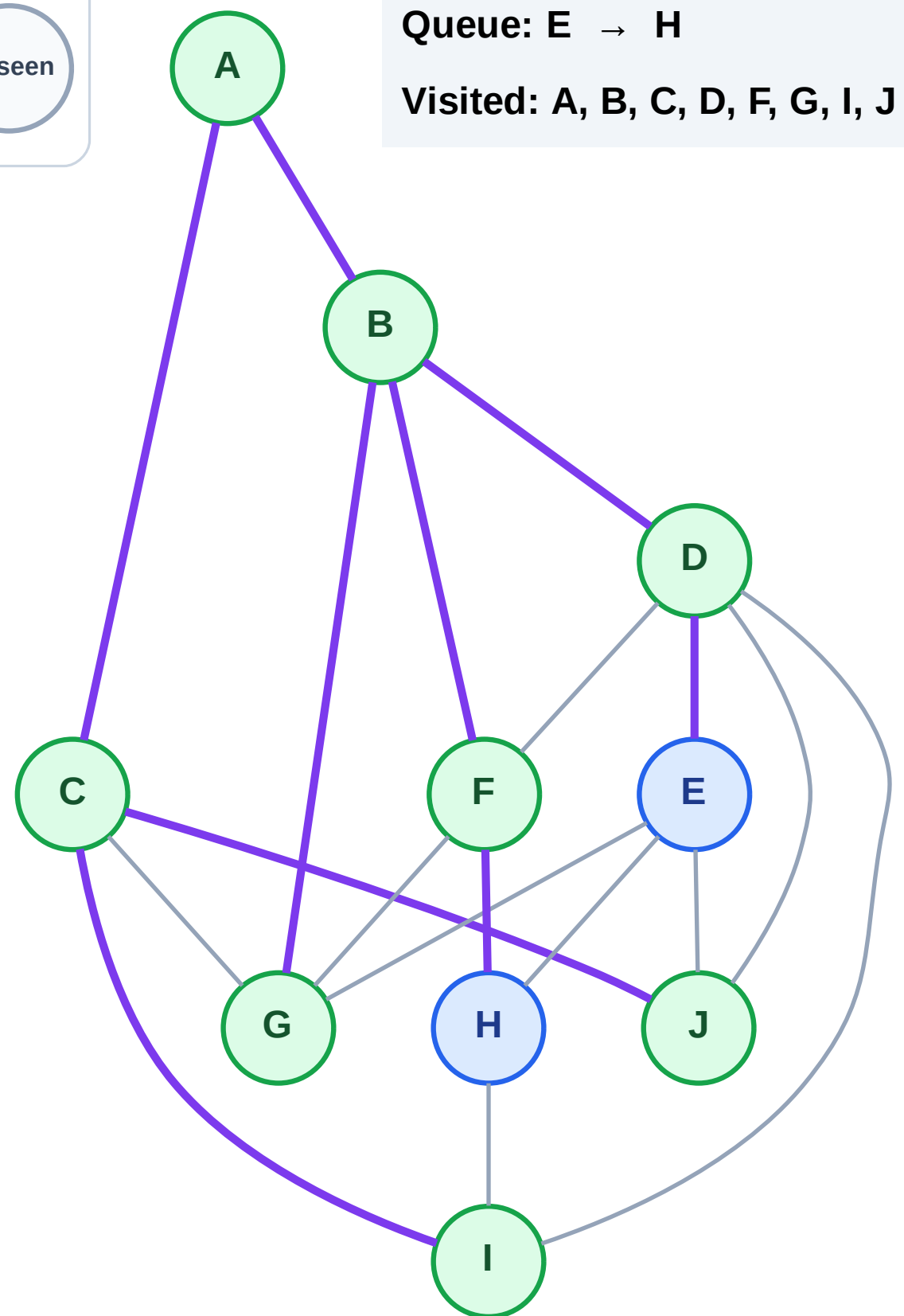
Step 17: No new neighbors from J

Legend



Queue: E → H

Visited: A, B, C, D, F, G, I, J



BFS Traversal

Step 18: Process node E

Legend

Complete

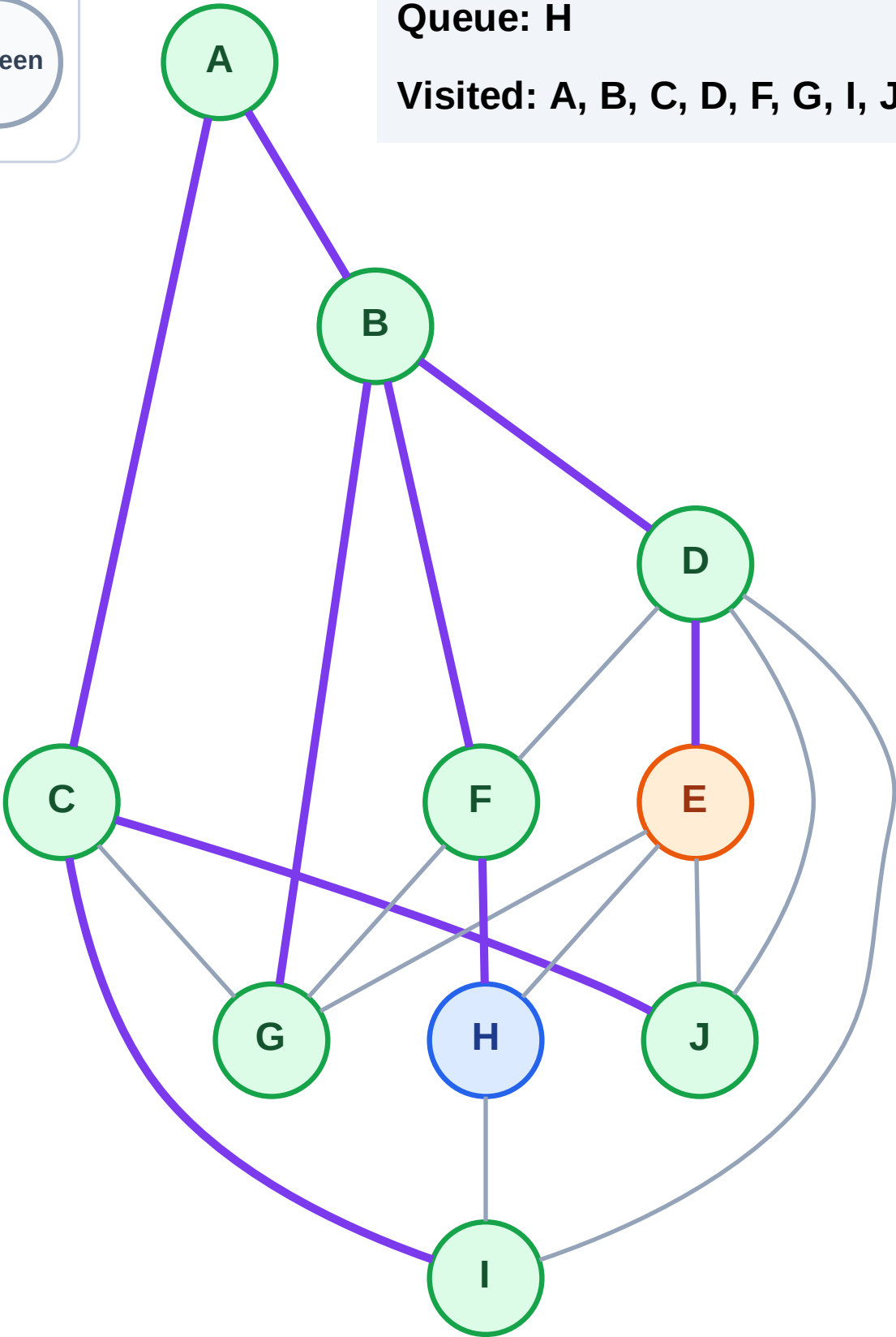
Processing

Discovered

Unseen

Queue: H

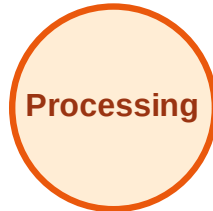
Visited: A, B, C, D, F, G, I, J



BFS Traversal

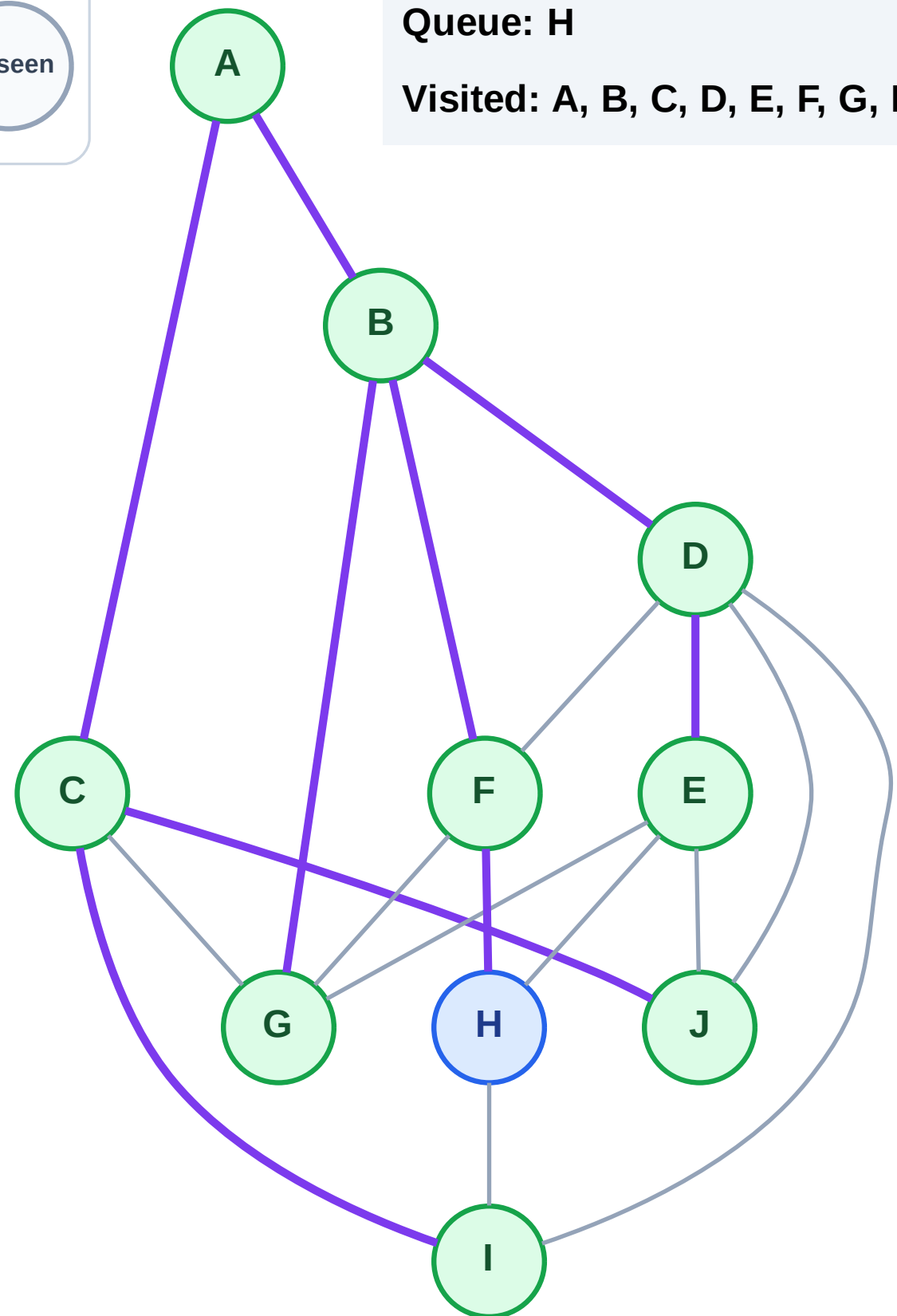
Step 19: No new neighbors from E

Legend



Queue: H

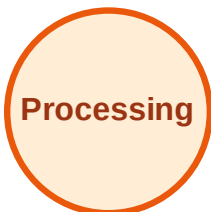
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

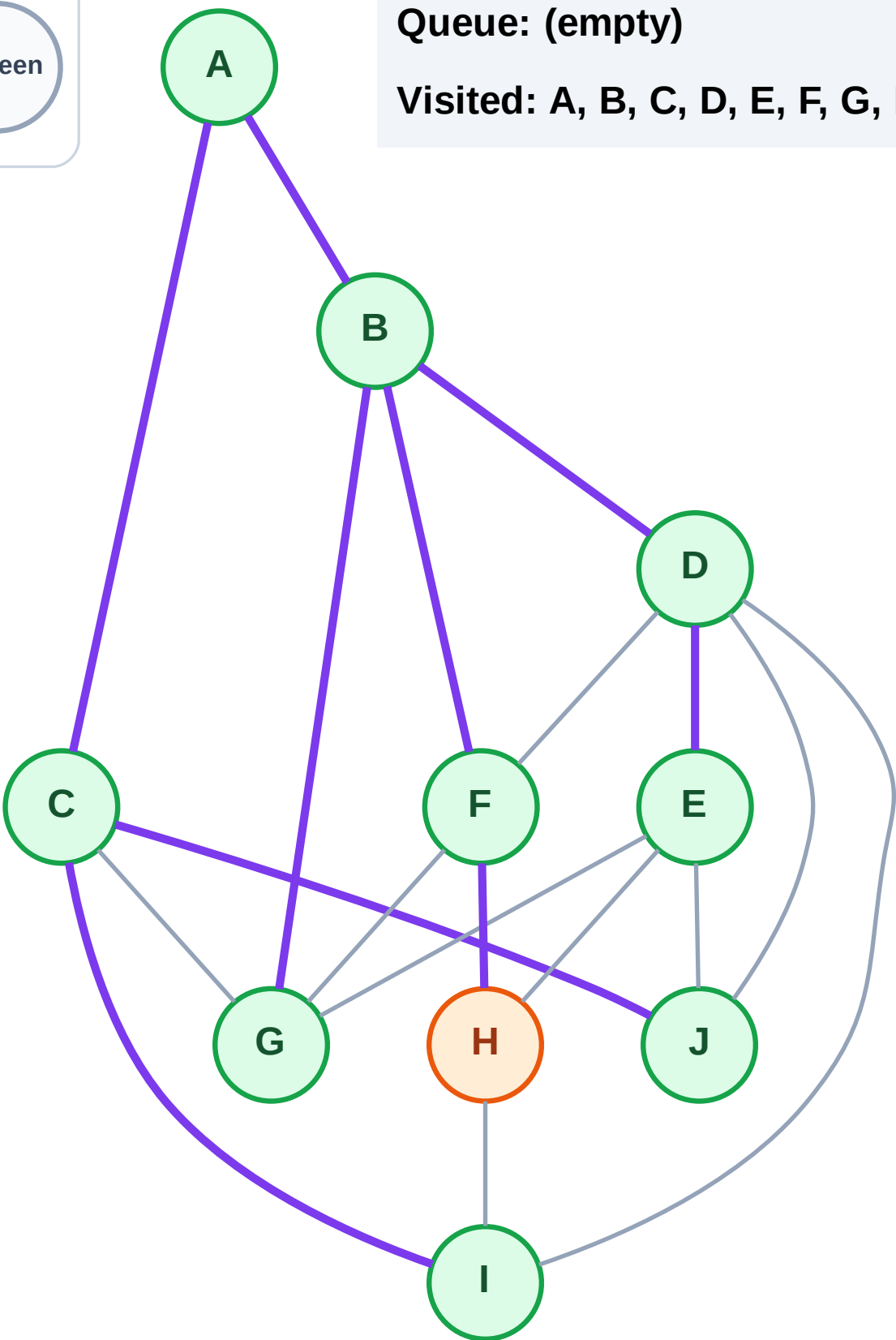
Step 20: Process node H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

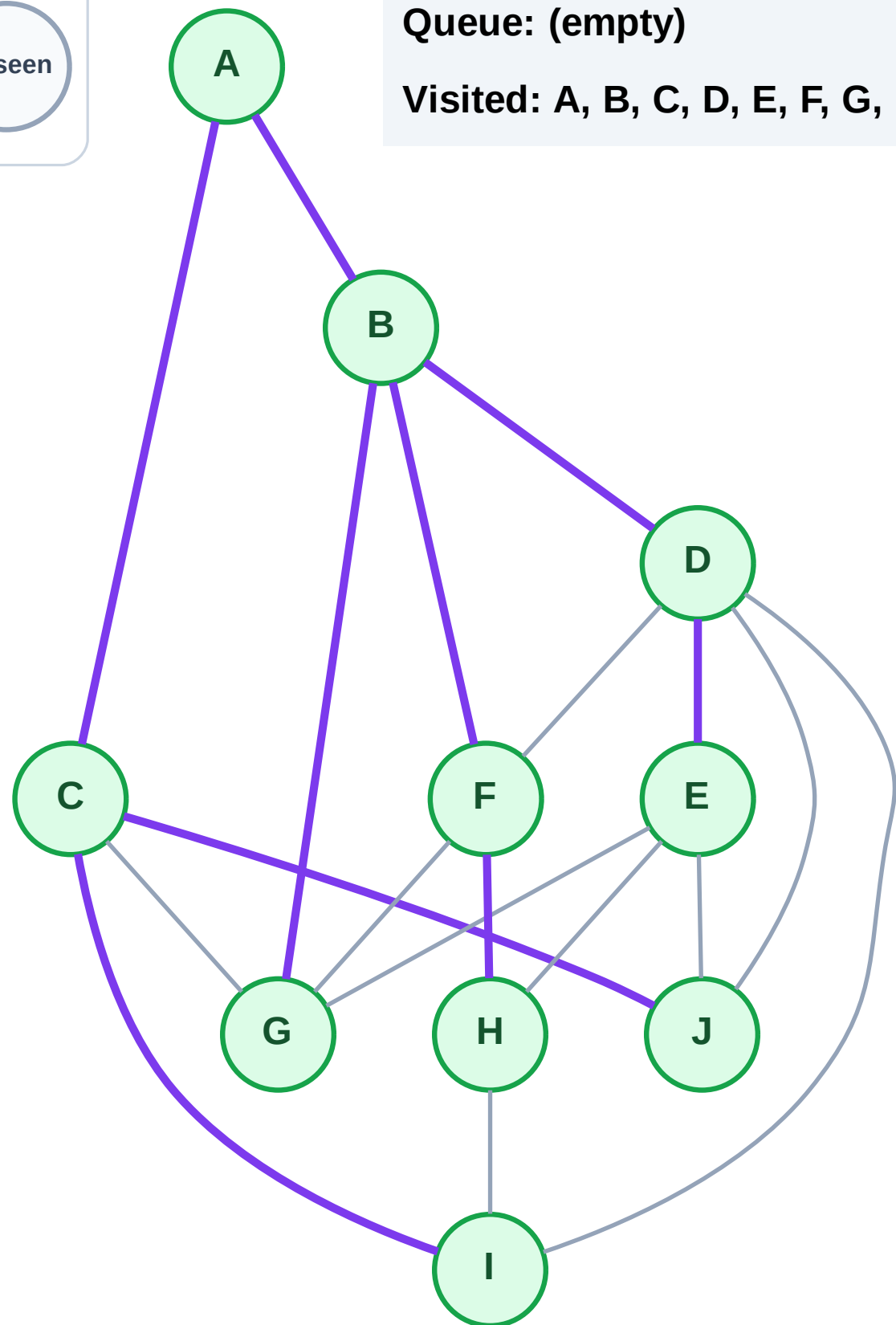
Step 21: No new neighbors from H

Legend



Queue: (empty)

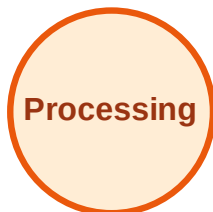
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

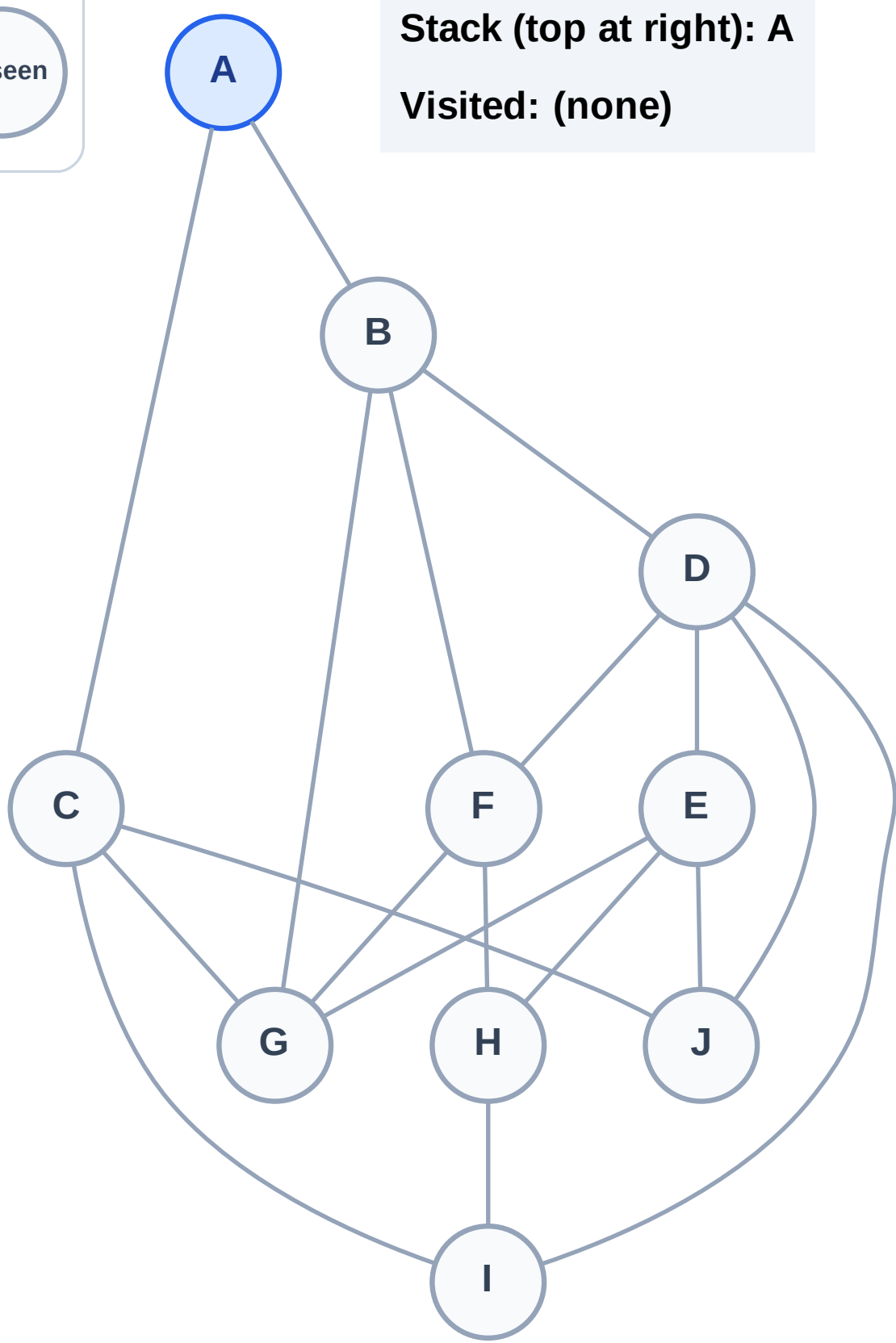
Step 1: Start at A

Legend



Stack (top at right): A

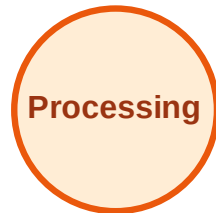
Visited: (none)



DFS Traversal

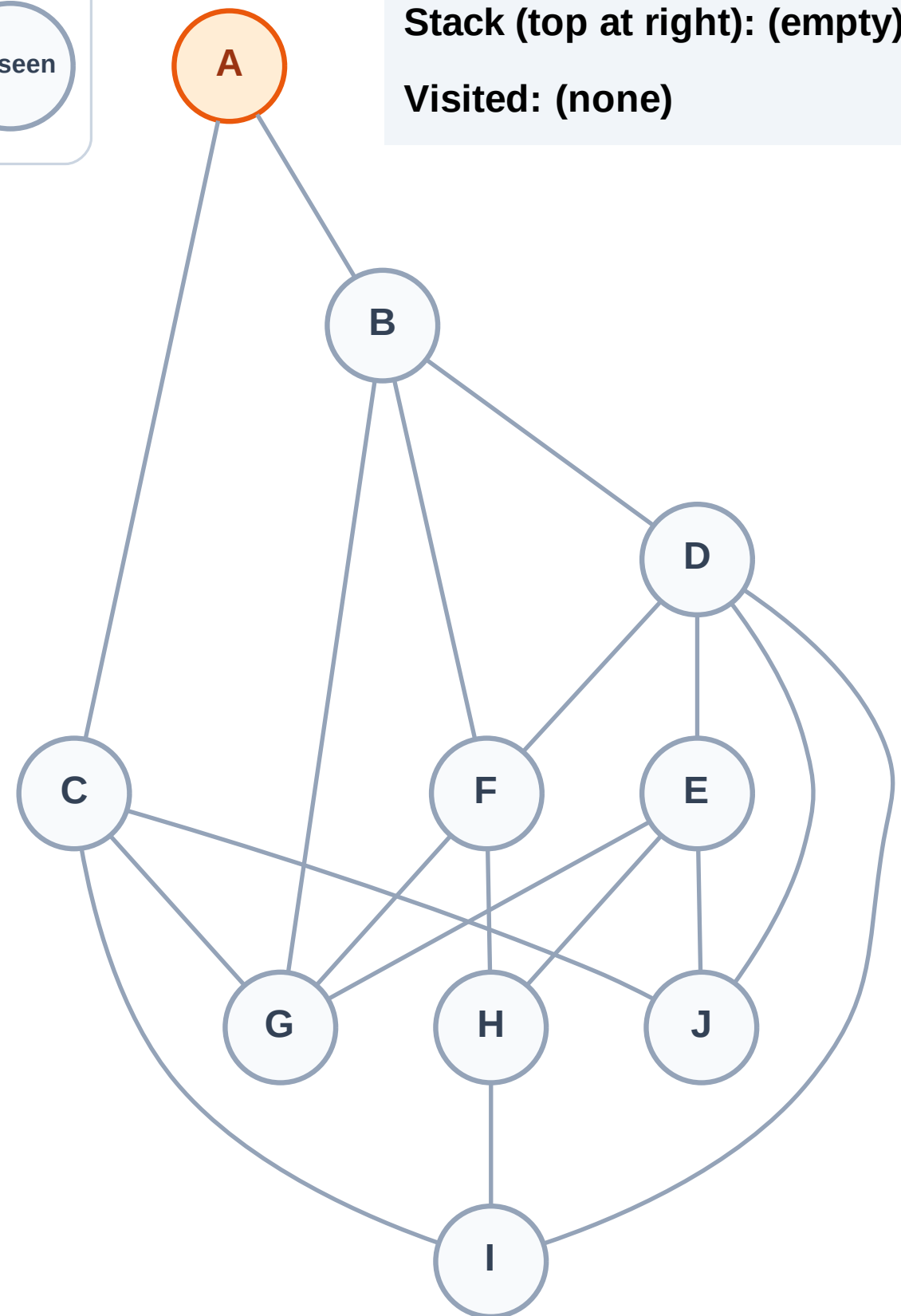
Step 2: Process node A

Legend



Stack (top at right): (empty)

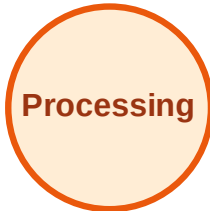
Visited: (none)



DFS Traversal

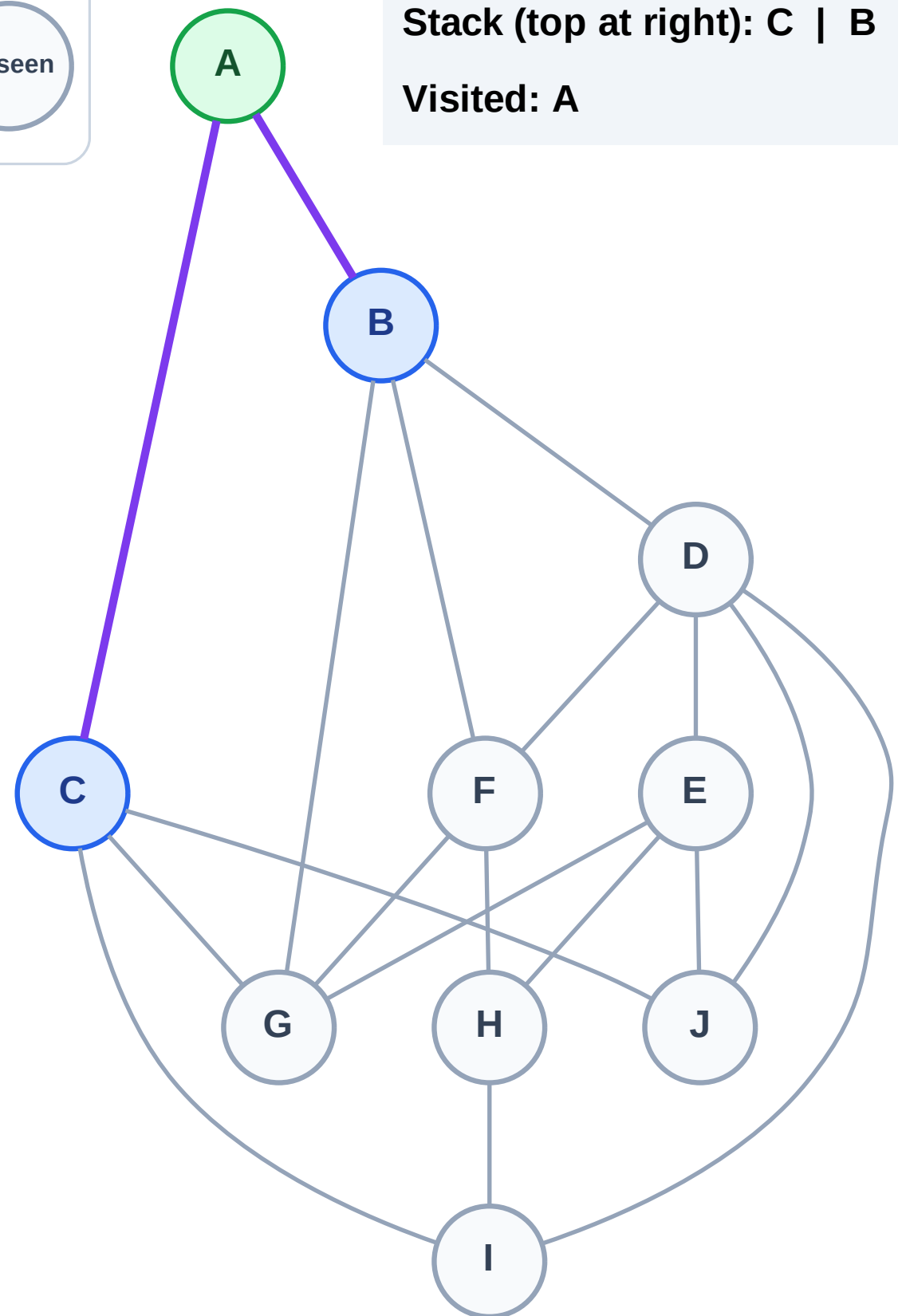
Step 3: Discover C, B from A

Legend



Stack (top at right): C | B

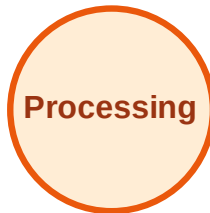
Visited: A



DFS Traversal

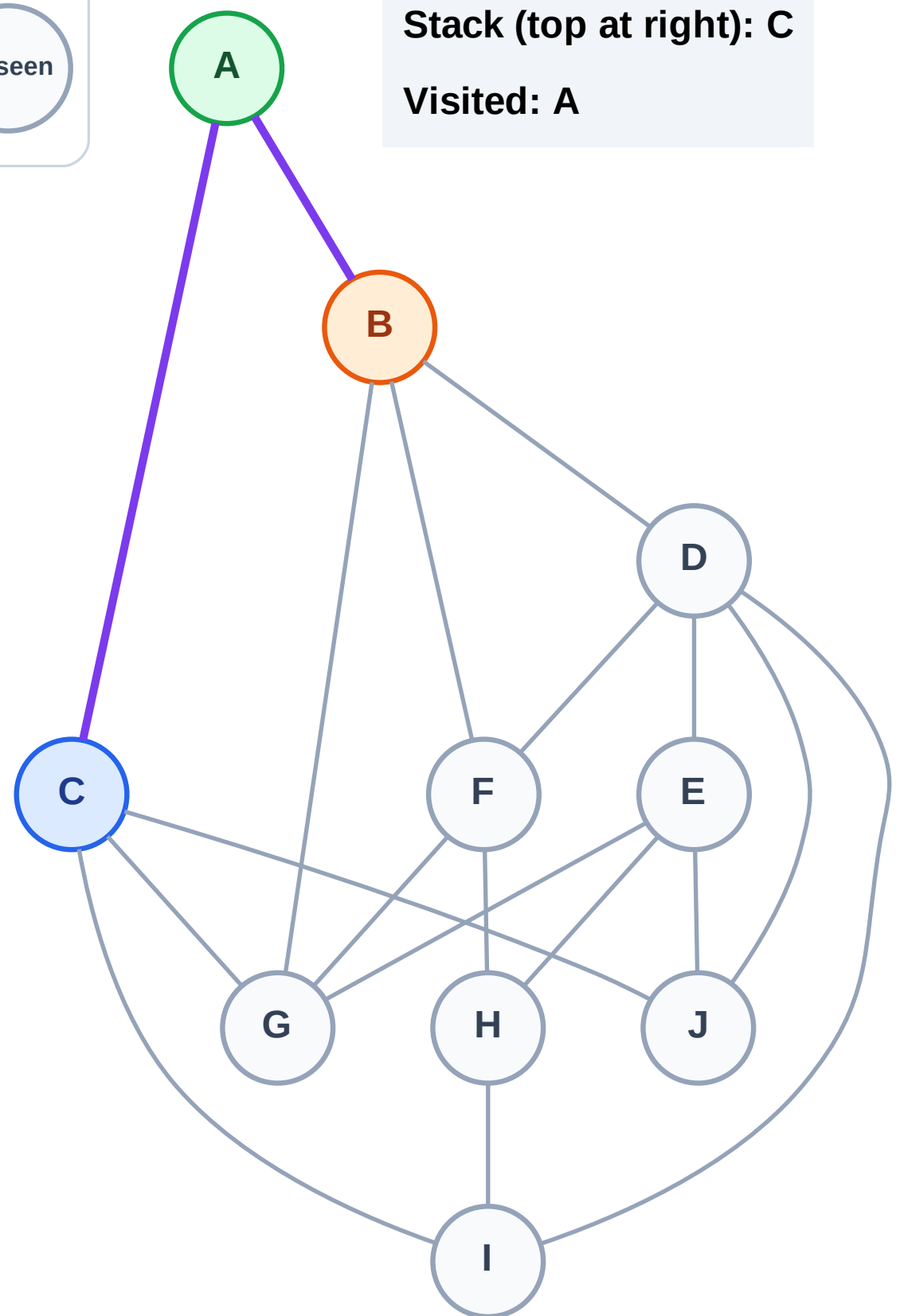
Step 4: Process node B

Legend



Stack (top at right): C

Visited: A



DFS Traversal

Step 5: Discover G, F, D from B

Legend

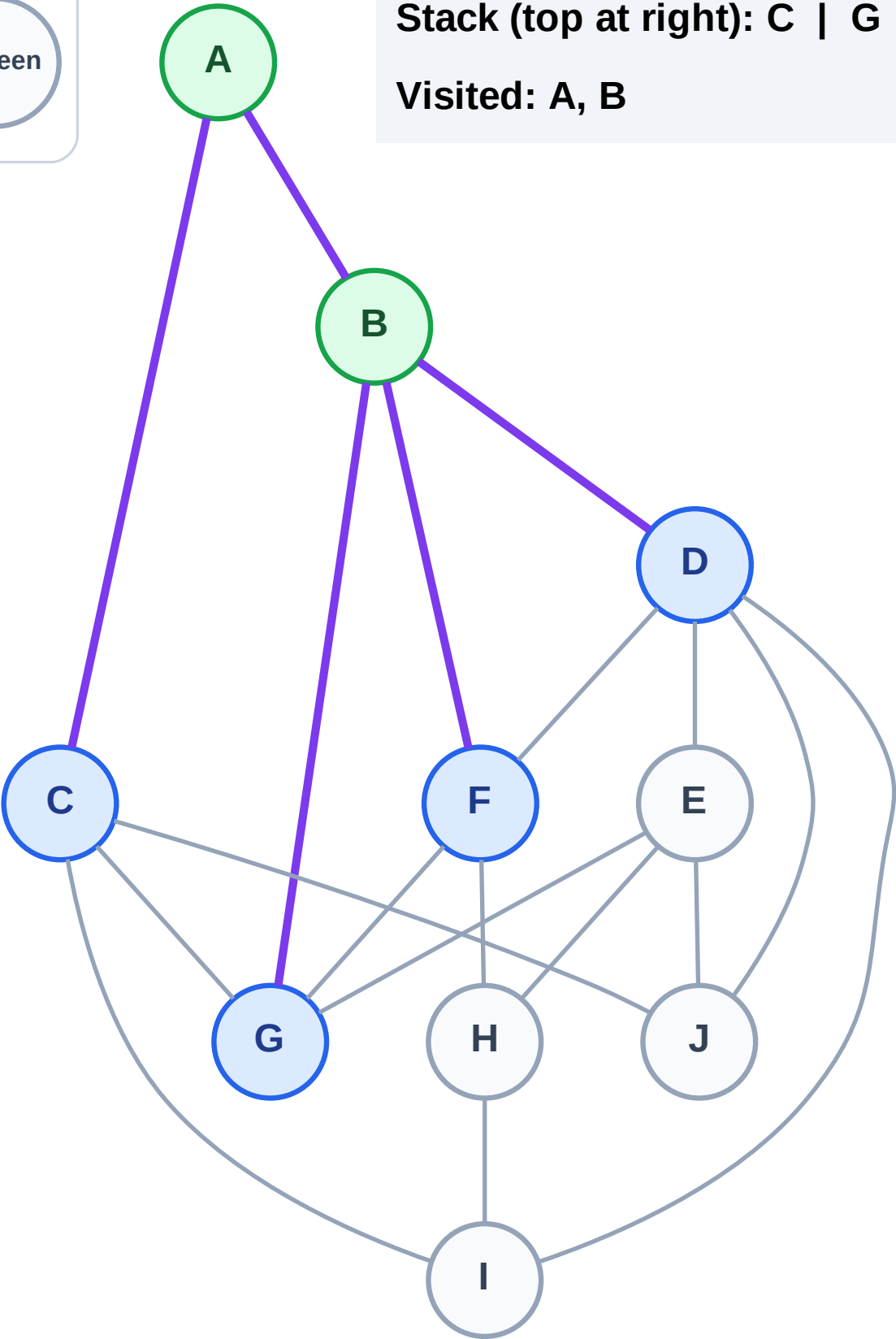
Complete

Processing

Discovered

Unseen

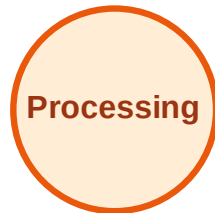
Stack (top at right): C | G | F | D
Visited: A, B



DFS Traversal

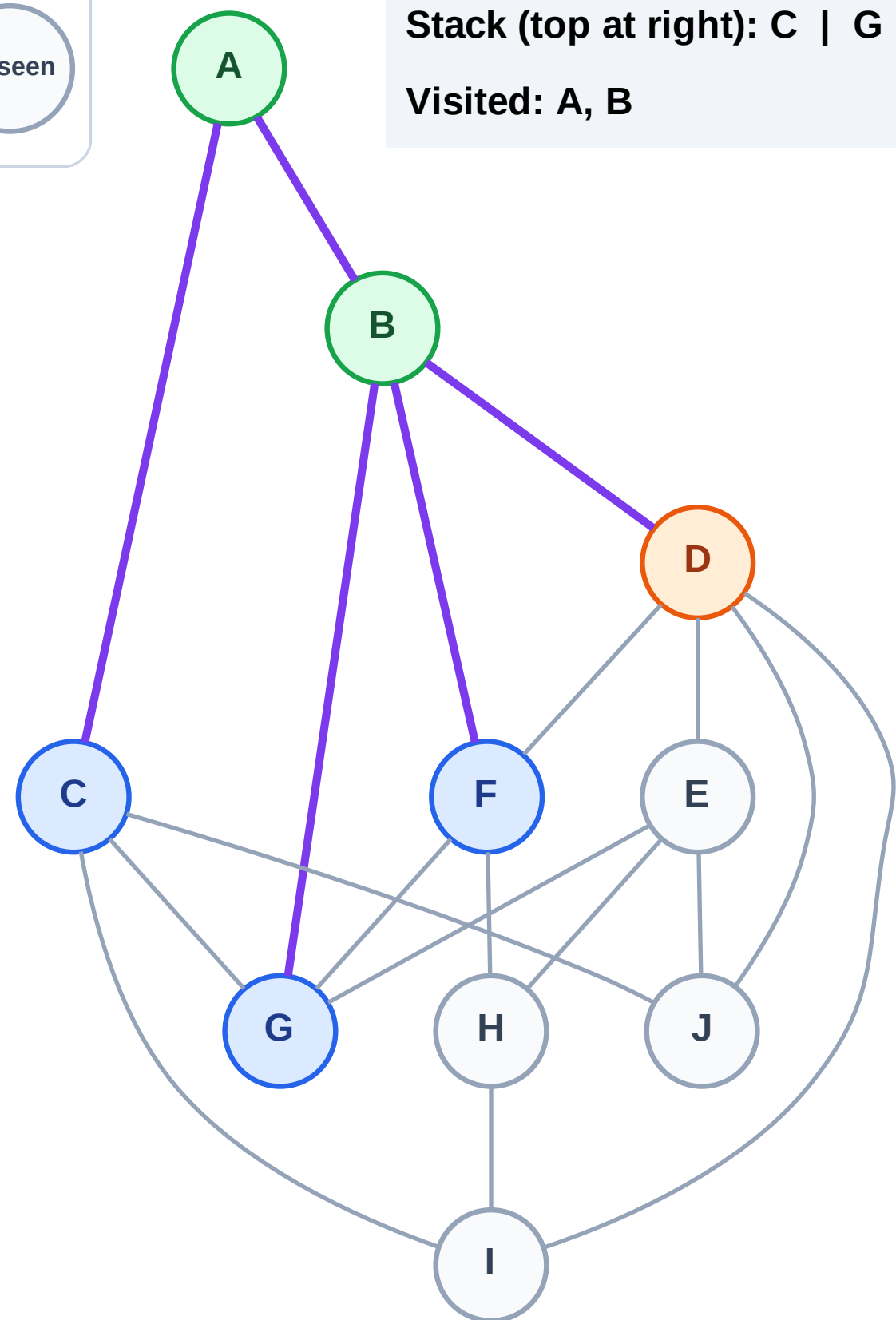
Step 6: Process node D

Legend



Stack (top at right): C | G | F

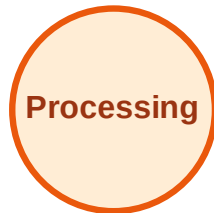
Visited: A, B



DFS Traversal

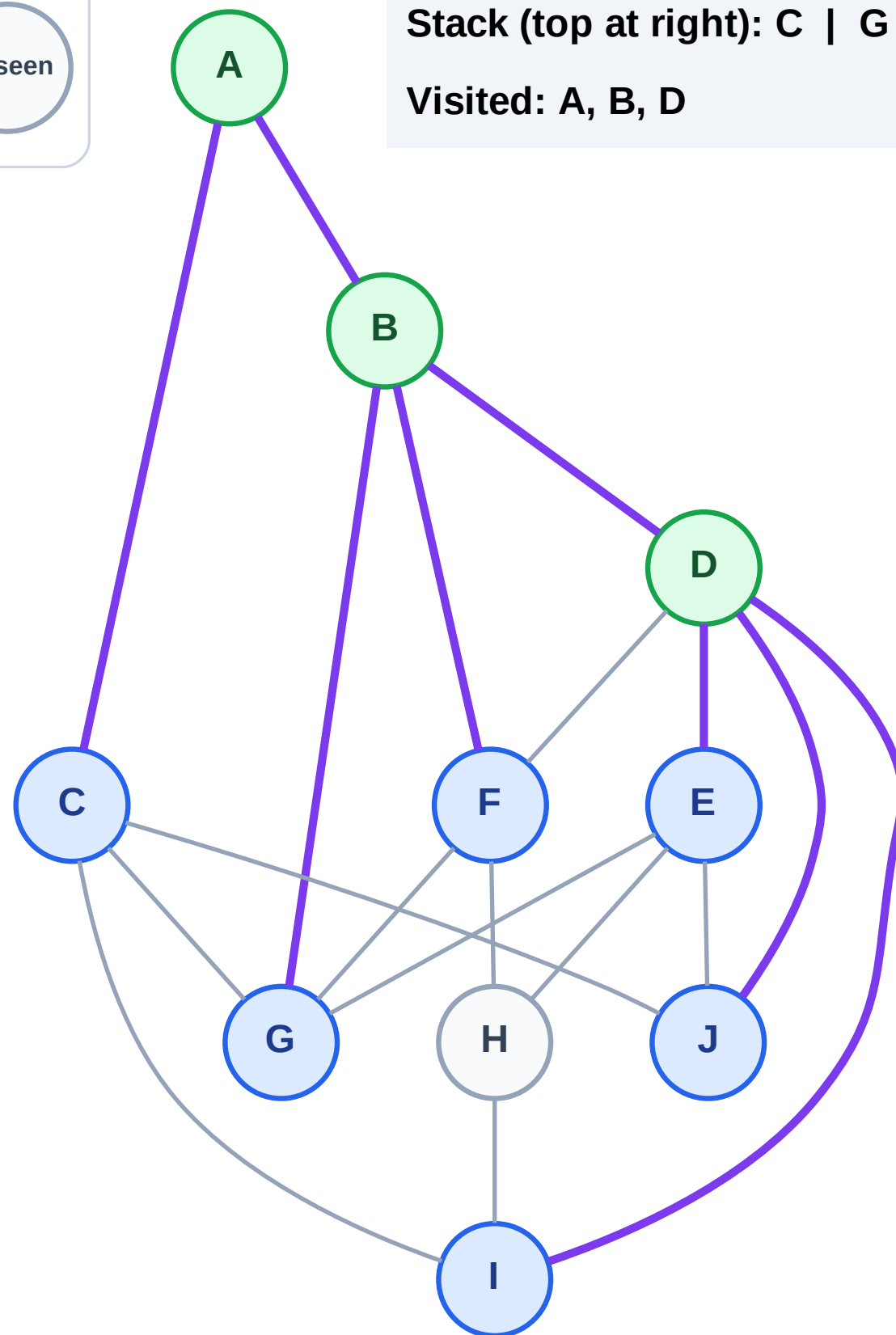
Step 7: Discover J, I, E from D

Legend



Stack (top at right): C | G | F | J | I | E

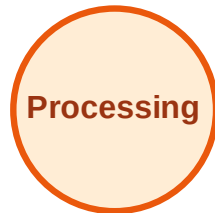
Visited: A, B, D



DFS Traversal

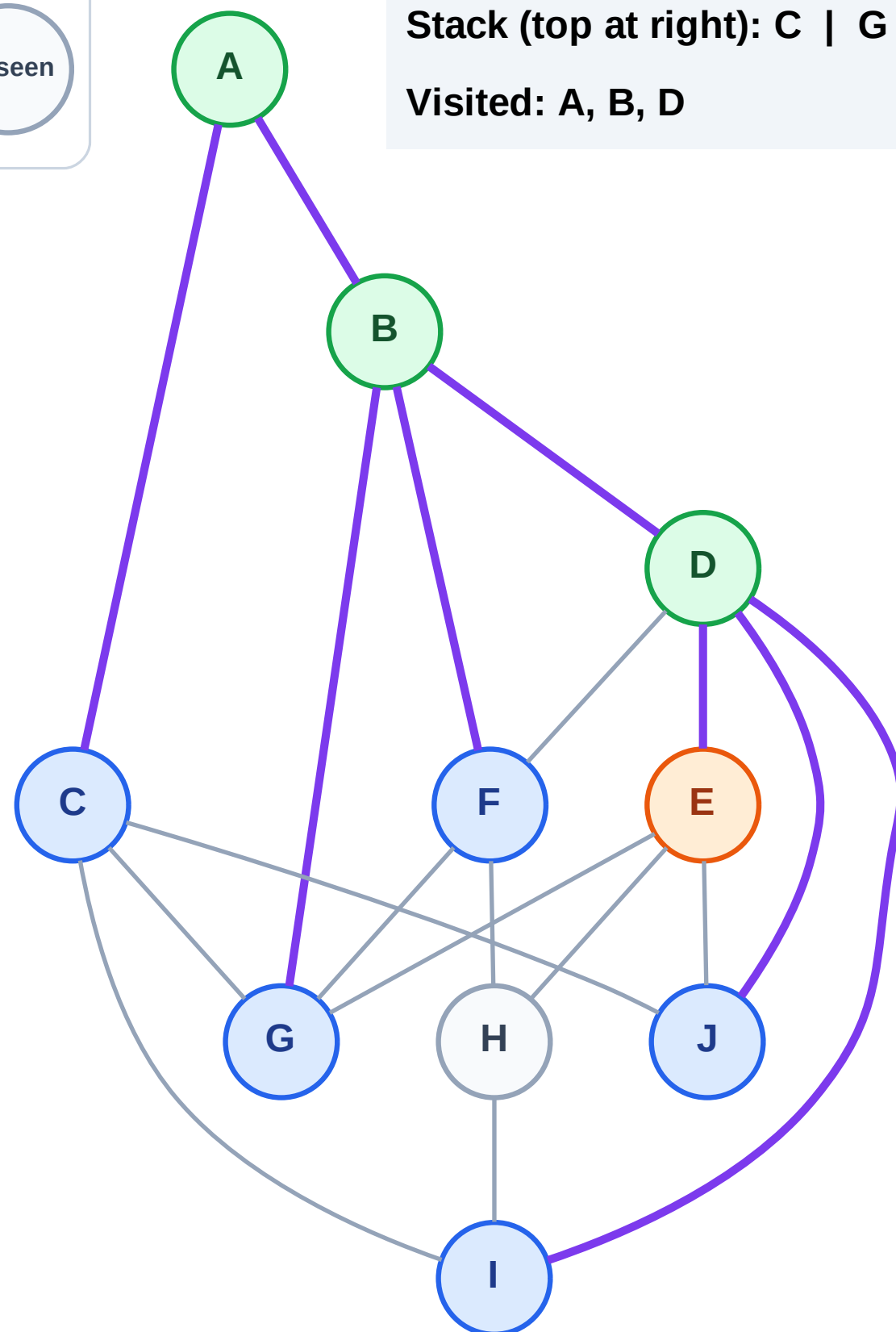
Step 8: Process node E

Legend



Stack (top at right): C | G | F | J | I

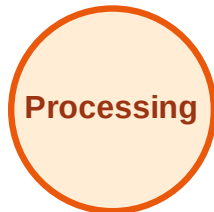
Visited: A, B, D



DFS Traversal

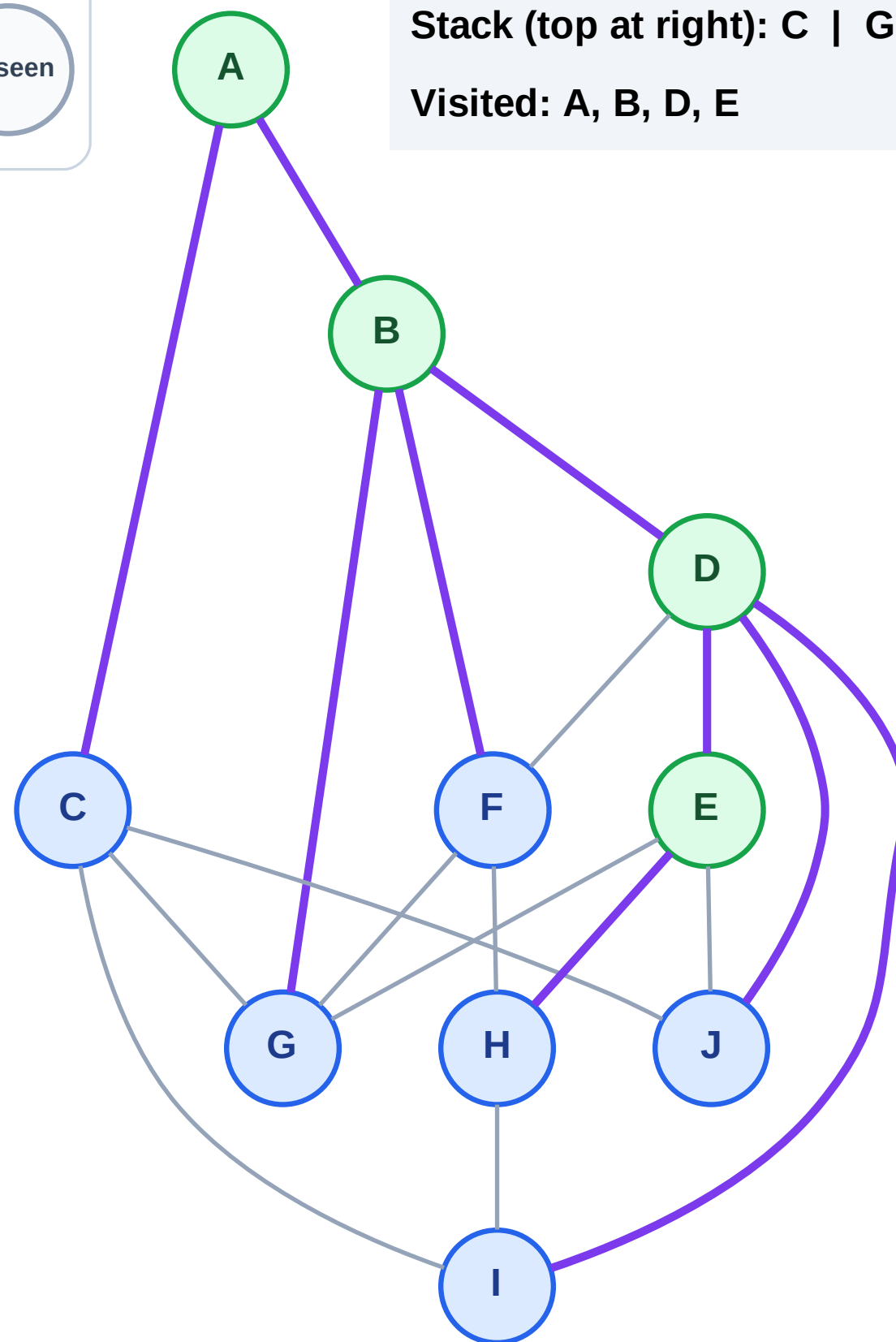
Step 9: Discover H from E

Legend



Stack (top at right): C | G | F | J | I | H

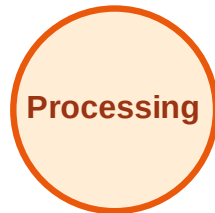
Visited: A, B, D, E



DFS Traversal

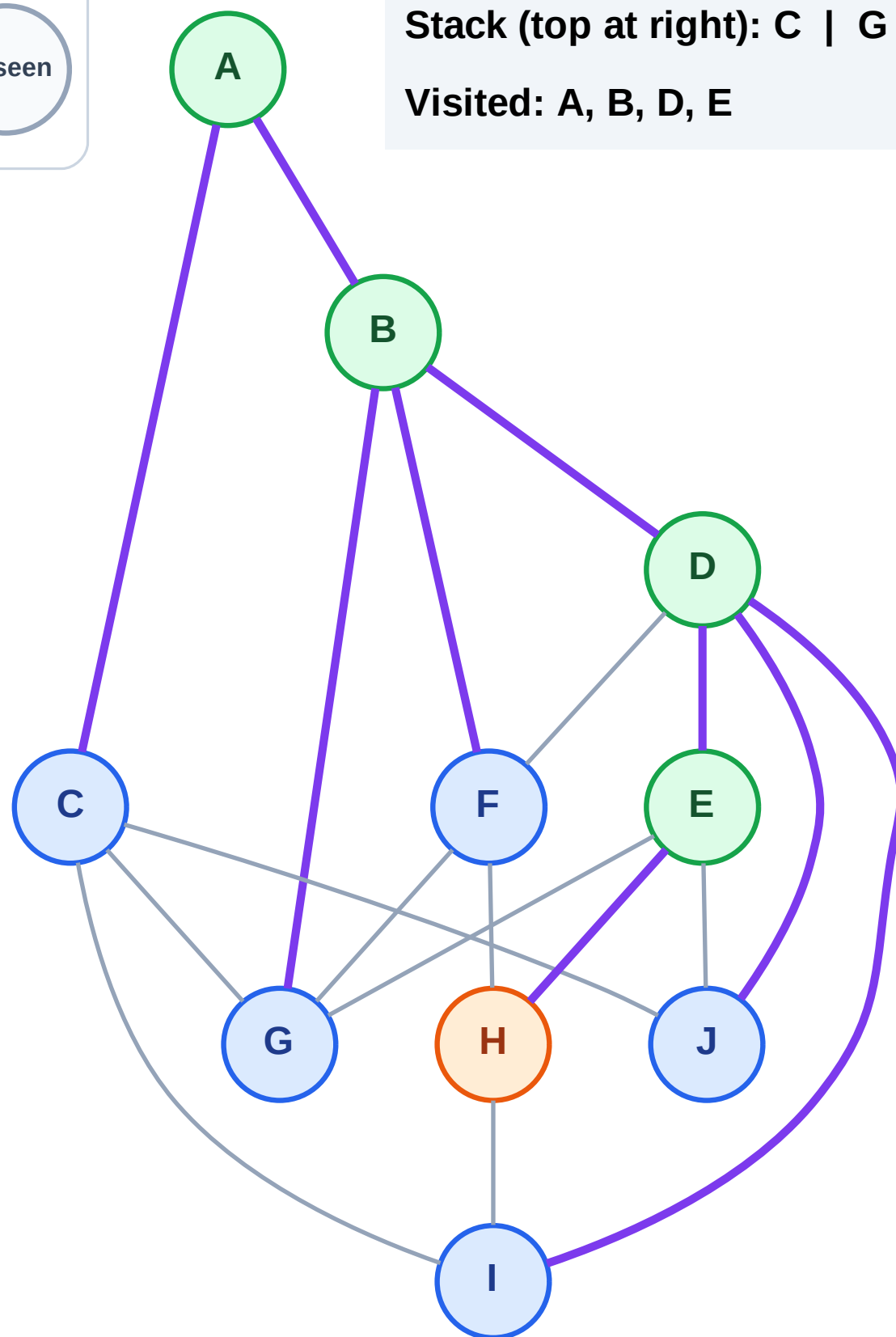
Step 10: Process node H

Legend



Stack (top at right): C | G | F | J | I

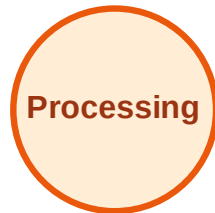
Visited: A, B, D, E



DFS Traversal

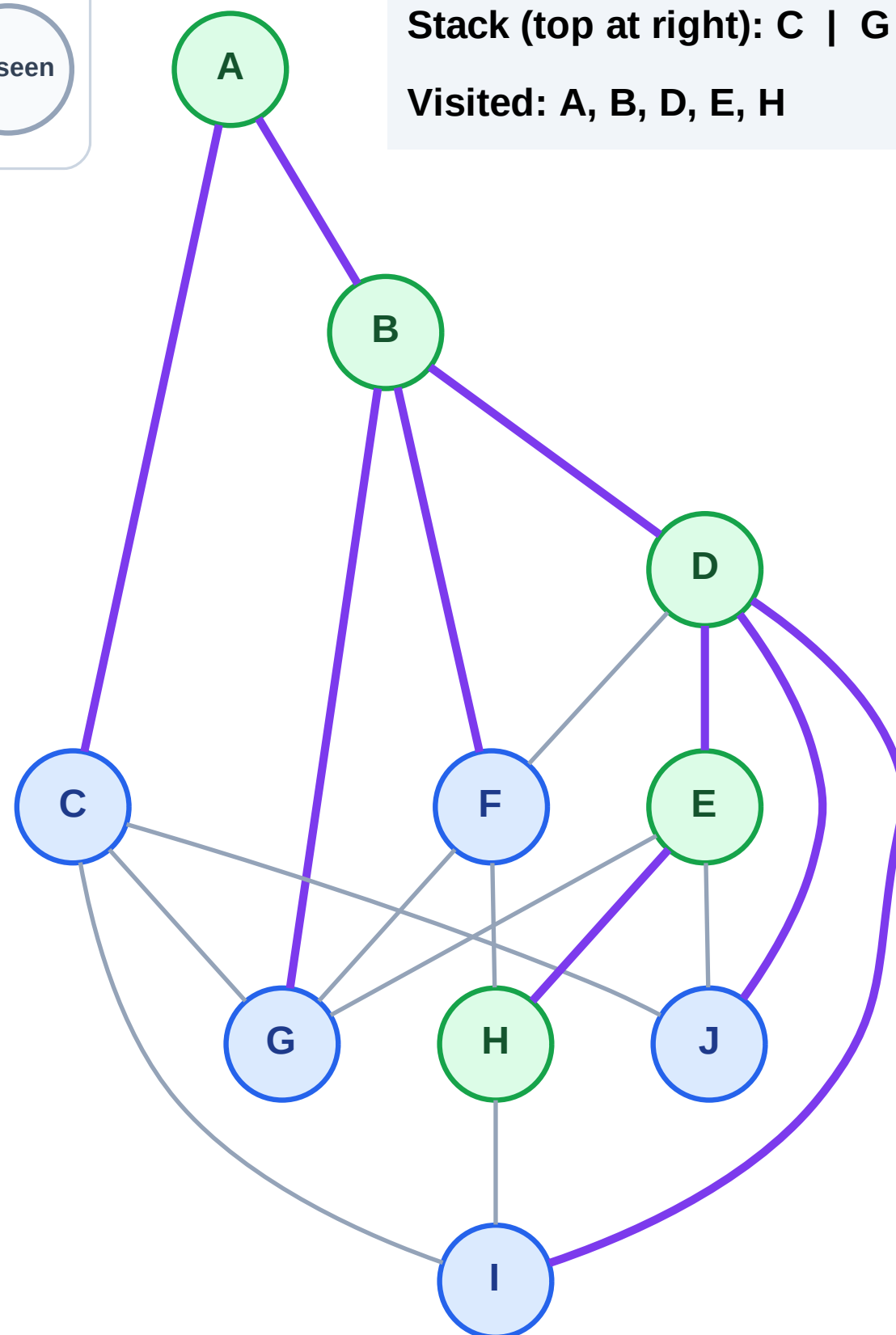
Step 11: No new neighbors from H

Legend



Stack (top at right): C | G | F | J | I

Visited: A, B, D, E, H



DFS Traversal

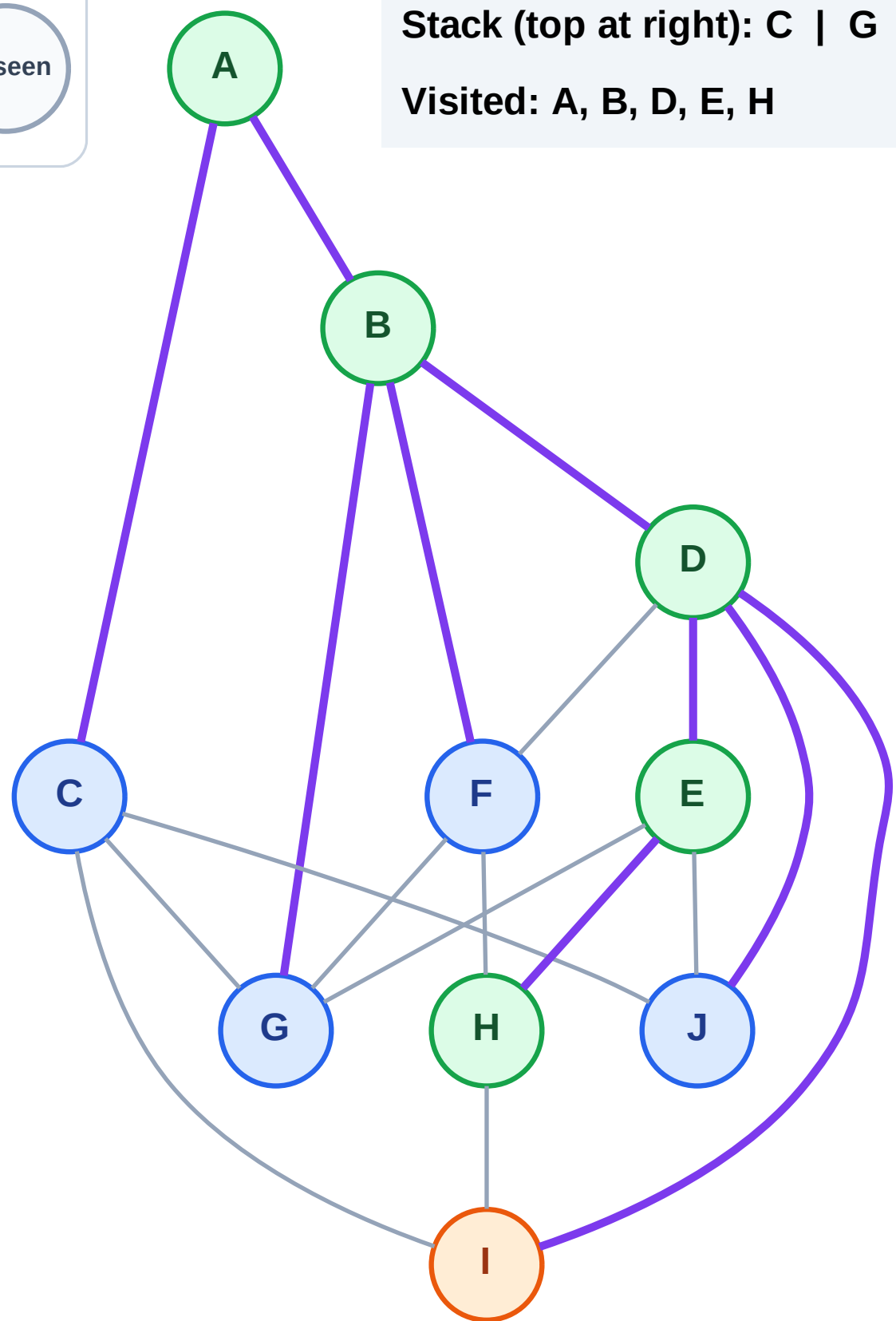
Step 12: Process node I

Legend



Stack (top at right): C | G | F | J

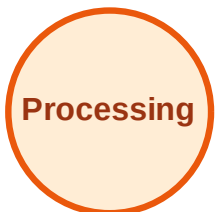
Visited: A, B, D, E, H



DFS Traversal

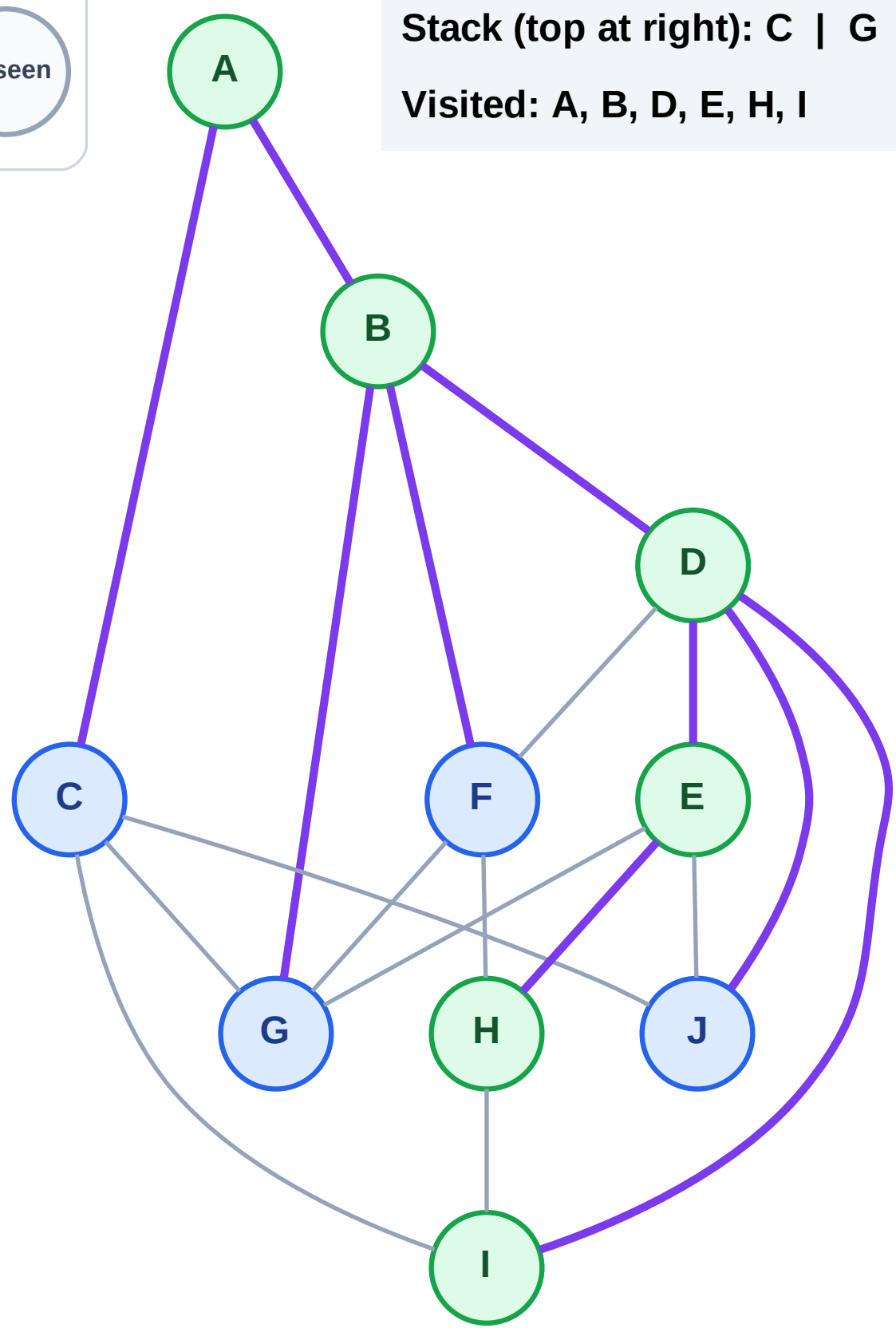
Step 13: No new neighbors from I

Legend



Stack (top at right): C | G | F | J

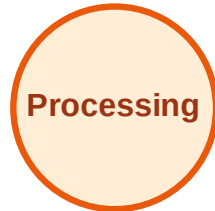
Visited: A, B, D, E, H, I



DFS Traversal

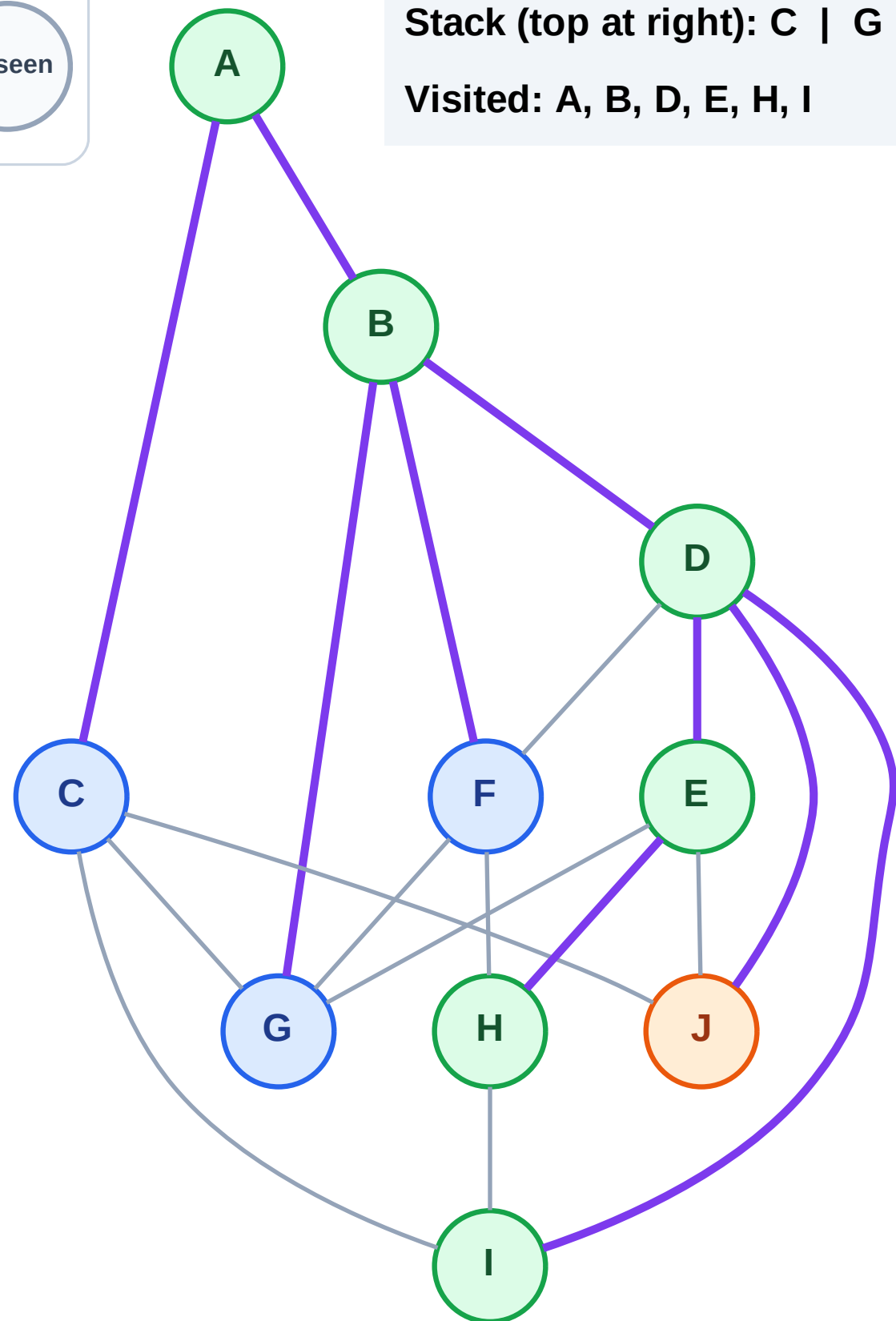
Step 14: Process node J

Legend



Stack (top at right): C | G | F

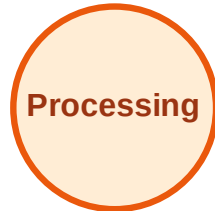
Visited: A, B, D, E, H, I



DFS Traversal

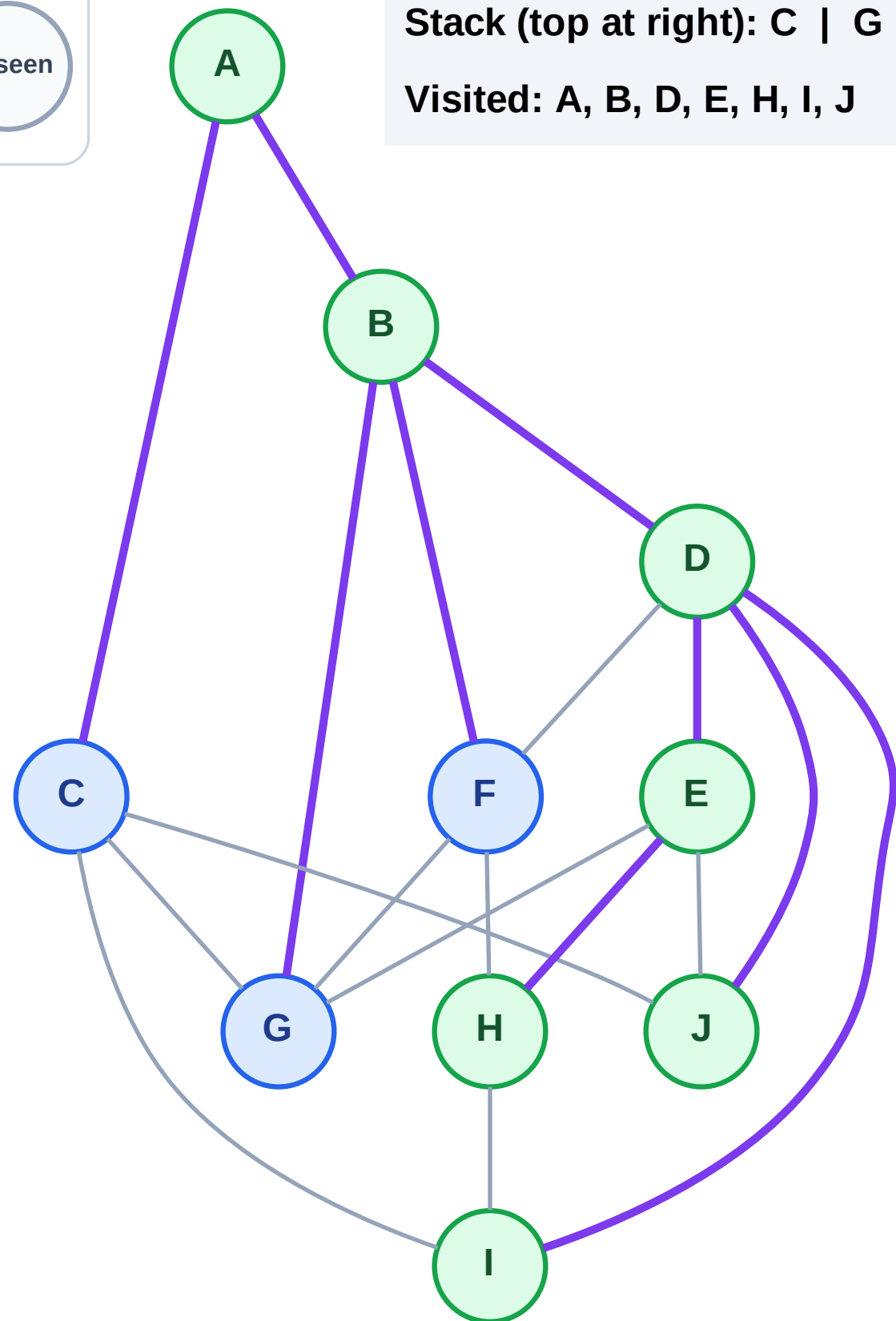
Step 15: No new neighbors from J

Legend



Stack (top at right): C | G | F

Visited: A, B, D, E, H, I, J



DFS Traversal

Step 16: Process node F

Legend

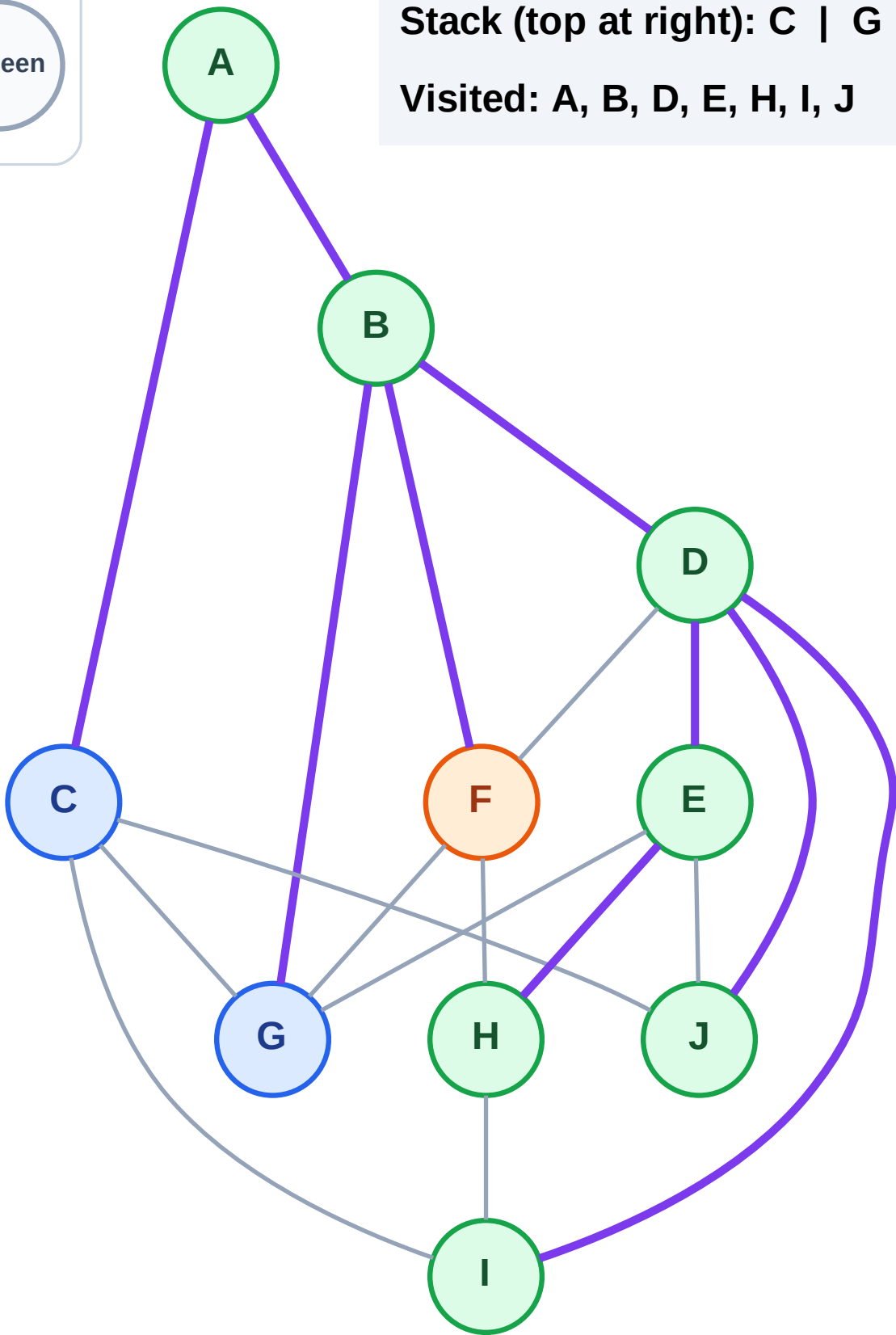
Complete

Processing

Discovered

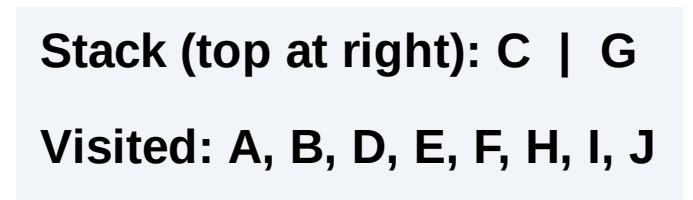
Unseen

Stack (top at right): C | G
Visited: A, B, D, E, H, I, J



Step 17: No new neighbors from F

Step 17: No new neighbors from F



DFS Traversal

Step 18: Process node G

Legend

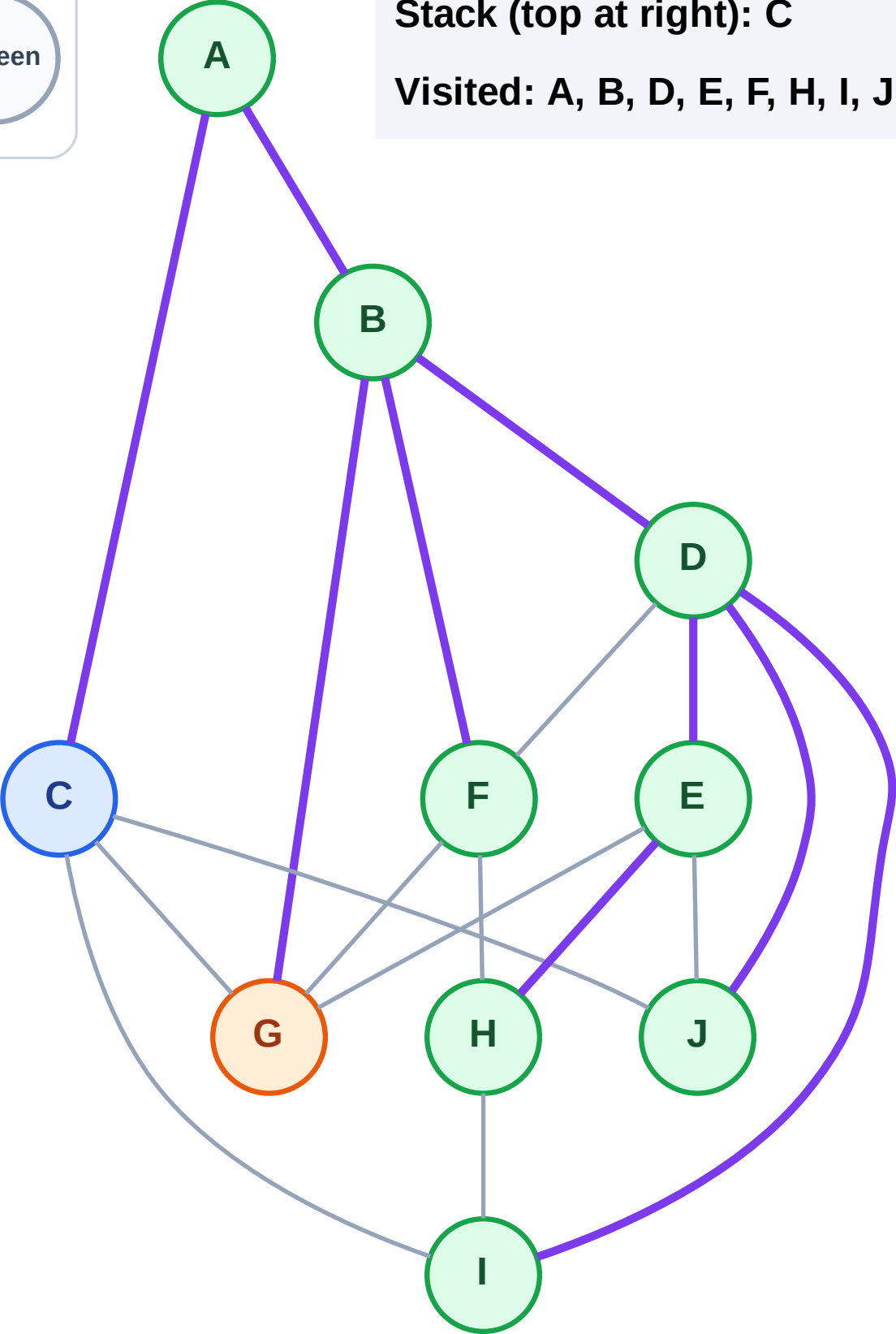
Complete

Processing

Discovered

Unseen

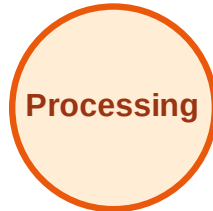
Stack (top at right): C
Visited: A, B, D, E, F, H, I, J



DFS Traversal

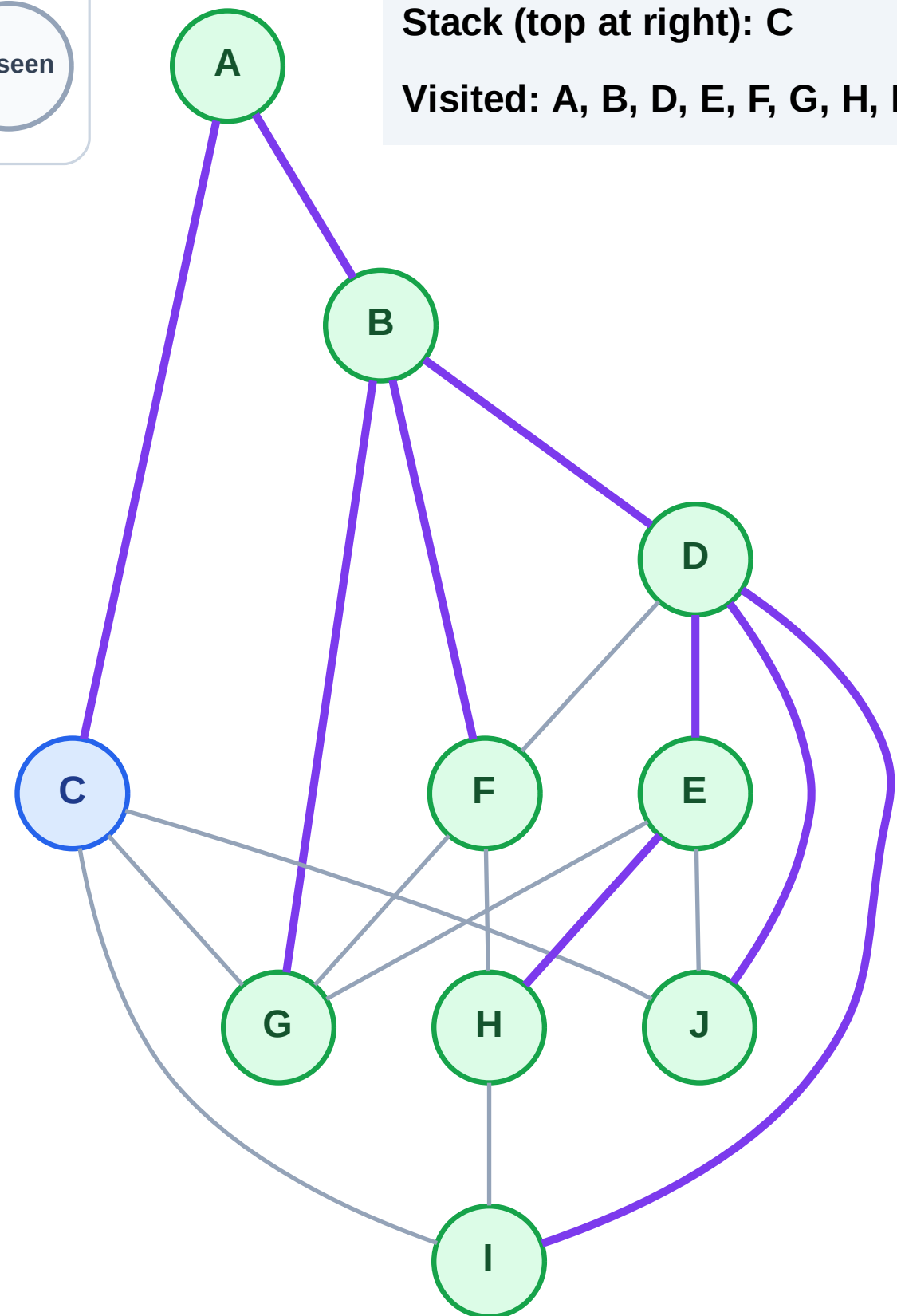
Step 19: No new neighbors from G

Legend



Stack (top at right): C

Visited: A, B, D, E, F, G, H, I, J

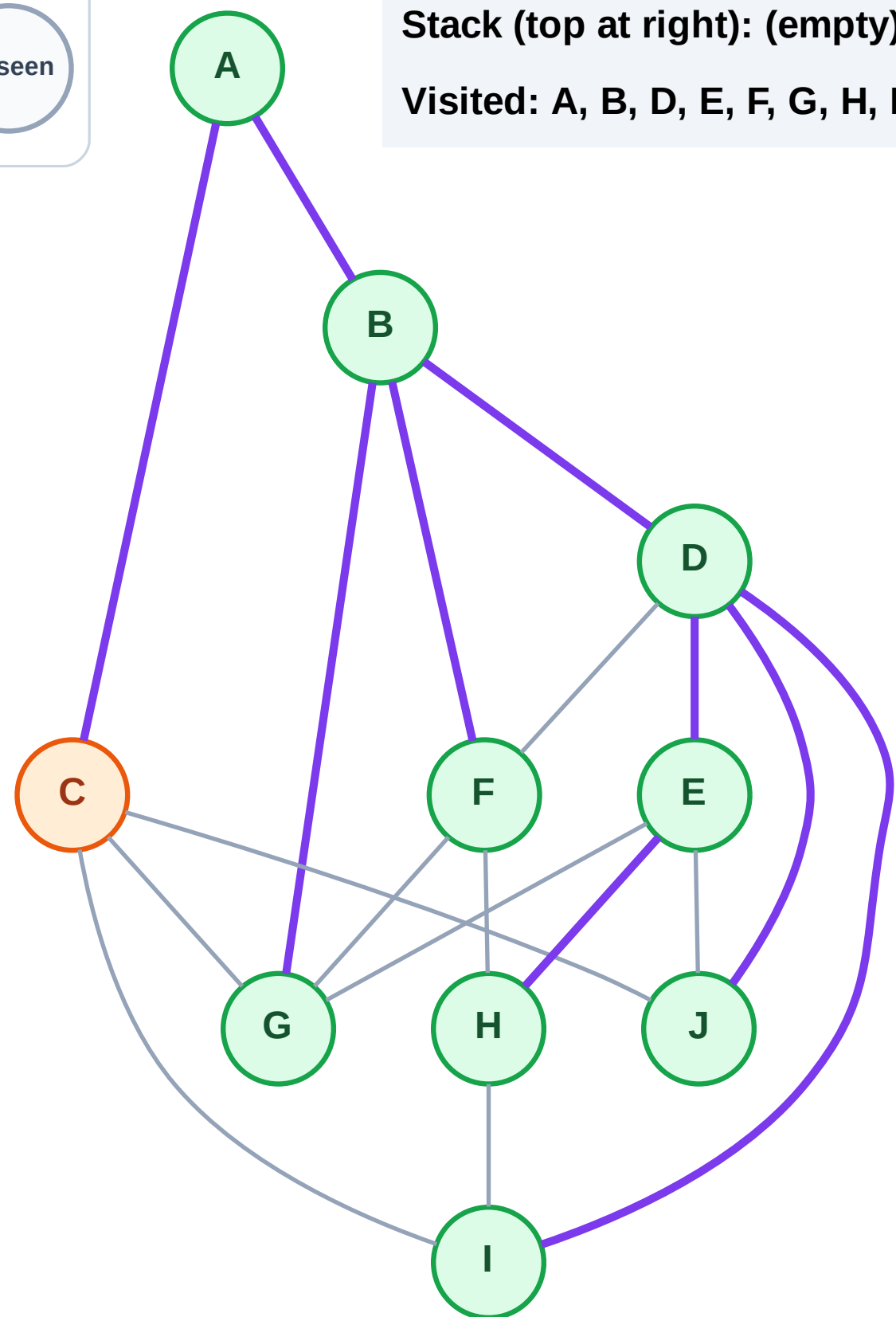


Step 20: Process node C

Legend

Unseen

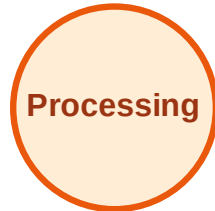
Visited: A, B, D, E, F, G, H, I, J



DFS Traversal

Step 21: No new neighbors from C

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

